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From a Single Demonstration to a General Policy for Contact-Rich Manipulation

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Abstract

We present a four-stage Learning from Demonstration (LfD) pipeline that incrementally constructs a general policy for multi-stage contact-rich manipulation from a single human demonstration. The pipeline is built around a crucial inductive bias—environmental constraints—that separates general task structure from hard-to-model instance-specific details. We show that, on top of this inductive bias, simple and well-established components are already sufficient to build an interactive LfD system that generalizes to unseen variations in object pose, local geometry, and contact conditions under real-world uncertainty. We validate the system on seven real-world multi-stage contact-rich manipulation tasks and achieve over 90% success across tasks. These results establish environmental constraints as fundamental building blocks for efficient generalization in learning from demonstration.

Keywords

learning from demonstration, perception-action coupling, compliance and impedance control, manipulation and grasping

1 Introduction

Learning from Demonstration (LfD) approaches compute policies from human demonstrations of task executions (Billard et al. 2008). The main objective in LfD is for these policies to be as general as possible, i.e., to successfully execute under substantial variations of the task instance. Achieving such generality from as few demonstrations as possible remains a fundamental challenge.

In this paper, we introduce an LfD method that generalizes from one human demonstration to unseen task variations. We test it on seven multi-stage, contact-rich manipulation tasks (see Fig. 1). Compared to prior work, it generalizes better while relying on a single demonstration, self-guided exploration, and a few targeted corrections rather than hundreds or thousands of demonstrations.

The proposed approach achieves such strong generalization because it carefully reasons about the structure of the problem, leverages insights about this structure, and uses them as inductive biases for generating and representing the produced policy. As we will see, the most important of these biases are based on environmental constraints (ECs) (Eppner et al. 2015). The general policy is then determined from a single demonstration based on four conceptual components, each described in more detail in Sec. 3.

First, information from the human demonstration is separated into a general part, which is assumed to be shared by all task instantiations, and an instance-specific part. Only the general part is used to produce the general policy. The required separation is achieved based on the inductive bias of ECs. These constraints capture general interaction patterns between the robot and its environment.

Second, the yet incomplete general policy is augmented by the robot by exploring local contact information along the general policy. This complements our understanding of ECs present in the task beyond what could be extracted from the demonstration. The result is an increase in the generality of the policy relative to the first step.

Third, so-called corrective demonstrations further increase the generality of the produced policy. Should the execution of the existing policy fail, more information from the human is required. The human kinesthetically corrects the failure, adding additional hard-to-model details to the policy, further increasing its generality.

Fourth, and finally, the robot fills in the details that vary across object instances using sensor feedback obtained from the interaction with ECs at run time. This step takes the general policy and makes it instance-specific again, leading to the successful generalization of the general policy.

Overall, our method produces general policies by separating out the general information from a demonstration, adding necessary details through augmentation and corrections, and adapting the policy during execution using sensory feedback. Through extensive real-world experiments, we demonstrate that this way of building policies from demonstrations leads to previously unmatched generality.

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Figure 1. Our LfD approach generalizes from a single demonstration and a few corrections more strongly than other approaches in the literature. This generalization manifests itself for a diverse set of multi-stage contact-rich manipulation tasks, including insertion puzzles, door locks with keys, coffee machine, drawers with handles, latch locks, Functional Manipulation Bench, and FurnitureBench. Key to such strong generalization is the use of environmental constraints as inductive biases.

2 Related Work

2.1 Environmental Constraint Exploitation

Since our main contribution lies in leveraging ECs to achieve generalization in LfD, we first review how prior work uses ECs in robot manipulation. Environmental Constraint Exploitation (ECE) refers to behaviors that exploit useful features of the environment to facilitate manipulation. This concept originates from studies showing that humans deliberately exploit contact to simplify grasp (Kazemi et al. 2014; Eppner et al. 2015) and assembly tasks (Klingbeil et al. 2017). While human studies have confirmed the value of ECE, its relevance to robotic manipulation has been recognized for decades. As early as the 1980s, Lozano-Pérez et al. showed that exploiting contact through compliant motions contributes to robustness under uncertainty in assembly tasks (Lozano-Pérez et al. 1984). Since then, contact exploitation has been applied across many areas, including grasping (Eppner and Brock 2015; Bonilla et al. 2014; Bimbo et al. 2019; Páll and Brock 2021), planning (Sieverling et al. 2017; Toussaint et al. 2014, 2018; Mao et al. 2023), robot co-design (Deimel et al. 2017; Vaish and Brock 2024), end-effector design (Catalano et al. 2014; Odhner et al. 2014; Deimel and Brock 2016; Puhlmann et al. 2022), extrinsic manipulation (Erdmann and Mason 1988; Hou et al. 2020; Ren et al. 2025), and in-hand manipulation (Dafle et al. 2014; Bhatt et al. 2021; Patidar et al. 2023). We incorporate ECE into our LfD framework by representing demonstrations as sequences of ECE behaviors.

2.2 One-Shot Imitation Learning

Recent advances in vision and language foundation models have driven growing interest in end-to-end imitation learning, where observations from demonstrations are mapped directly to actions. Although these methods have been successfully applied to a variety of manipulation tasks, they are typically data-intensive and often require hundreds or thousands of demonstrations per task (Brohan et al. 2023; Zitkovich et al. 2023; O'Neill et al. 2024; Khazatsky et al. 2024; Bjorck et al. 2025; Gemini Robotics Team et al. 2025; TRI LBM Team 2025; AgiBot-World-Contributors et al. 2025).

An alternative direction is one-shot imitation learning, which aims to learn a task from a single demonstration. Many approaches in this category formulate LfD as a pose estimation problem using object-centric representations such as object keypoints (Xue et al. 2023; Cai et al. 2024b; Di Palo and Johns 2024; Allu et al. 2025; Ding et al. 2025; Chen et al. 2025a; Wichitwechkarn et al. 2025; Tang et al. 2025a) or poses (Lee et al. 2020; Vitiello et al. 2023; Heppert et al. 2024; Allu et al. 2025). A demonstrated behavior can then be reproduced by executing the trajectory defined relative to the task object. Since object-centric representations abstract away task-irrelevant details like visual descriptors and object poses, these methods generalize well to such variations. When combined with advanced vision techniques, recent work has shown that this learning paradigm allows the acquisition of thousands of manipulation behaviors from only tens of hours of demonstration data (Dreczkowski et al. 2025).

However, within this object-centric, trajectory-replay formulation, actions are typically represented as explicit Cartesian trajectories relative to the object (Lee et al. 2020; Vitiello et al. 2023; Heppert et al. 2024). In the context of contact-rich manipulation, this representation inherently overfits to specific object geometries; a policy learned from a particular latch or door handle rarely generalizes to geometrically distinct instances. Furthermore, tracking Cartesian trajectories is highly susceptible to failure due to unmodeled contact dynamics, such as slippage. To address these limitations, we introduce a novel action representation based on ECs. Building on this representation, our approach achieves superior generalization across varying object geometries and significantly enhances robustness against positional uncertainties.

2.3 Learning With Manipulation Primitives

Designing manipulation primitives enables the injection of inductive biases that capture task-relevant structure and thereby improve generalization to task-irrelevant variations (Felip et al. 2013; Kroemer et al. 2021). This idea has been used to learn contact-rich manipulation skills such as object alignment (Skubic and Volz 2000), part assembly (Suomalainen et al. 2021; Hu et al. 2024), and long-horizon extrinsic manipulation (Wu et al. 2024).

Our approach follows this direction by representing a demonstration as a sequence of parameterized primitives that exploit ECs.

The most notable distinction between our framework and prior work is that we do not attempt to extract fully specified primitives from demonstration. Rather, we deliberately leave instance-specific parameters under-specified, allowing them to be continuously updated online via sensory feedback. This online adaptation fills in instance-specific details that differ from the demonstration and accounts for uncertainties, leading to strong generalization in real-world manipulation tasks.

2.4 Learning Geometric Constraints From Demonstrations

Since our EC primitives are built on geometric constraint models, our work is closely related to research on extracting such constraints from demonstrations (Dang and Allen 2010; Niekum et al. 2015a; Pérez-D'Arpino and Shah 2017; Liu et al. 2019; Jain and Niekum 2020; Subramani et al. 2018; Wake et al. 2021; Li and Brock 2022; Hegeler et al. 2025; Gao et al. 2023; Hu et al. 2024; Pfisterer et al. 2025; Mohammadi et al. 2026; Overbeek et al. 2025a,b). These approaches typically fit geometric models, e.g., prismatic or revolute joints, to end-effector trajectories. However, this mapping is often ambiguous, as a single trajectory may be consistent with multiple constraint types.

Some prior approaches attempt to address this ambiguity by incorporating force information (Subramani et al. 2018; Overbeek et al. 2025a). However, as noted in (Li et al. 2023b; Suomalainen et al. 2022), human demonstrations could lack the necessary contact information, especially as demonstrators become familiar with the task and their motions become more exploitative. Our approach addresses this by augmenting the demonstration with active exploration that generates additional contact information for the disambiguation. While the idea of learning constraints with active exploration has been explored in (Hausman et al. 2015; Ortenzi et al. 2016; Lin and Mistry 2020), they typically assume that each task is covered by a single constraint. In contrast, we consider multi-stage manipulation tasks involving sequential constraints. We also present a method to compute the most informative exploratory actions to efficiently disambiguate constraint candidates.

2.5 Interactive Learning From Demonstration

Finally, we summarize research on interactive learning from demonstration. This line of research has received growing attention as a way to improve policy robustness through human-in-the-loop corrections (Ross et al. 2011; Celemin et al. 2022). A critical prerequisite for such correction is failure detection. Existing approaches often rely on out-of-distribution detection (Nemec et al. 2020; Takeuchi et al. 2023; Liu et al. 2024; Vanc et al. 2025) or supervised classifiers trained to distinguish between nominal and failure states (Kappler et al. 2015; Niekum et al. 2015b). In contrast, our method detects failures by monitoring contact transitions extracted during a self-guided exploration phase. Deviations from expected transitions signal that a manipulation primitive has terminated incorrectly.

Once a failure is detected, we incorporate human corrections to recover. Unlike prior methods that represent corrections as Cartesian trajectories guiding the robot to particular poses (Ross et al. 2011; Niekum et al. 2015b), we model corrections as behaviors that bring the robot back to a nominal contact state, from which it can continue to exploit ECs. This idea is conceptually similar to (Xiang et al. 2024), where human input guides the robot back to a safe state to avoid premature termination. By expressing corrections at the level of EC-based abstractions rather than low-level actions, we further promote generalization across task variations.

3 From A Single Demonstration To A General Policy

The core challenge in LfD is generalization. To produce a general policy, we must separate generalizable information of demonstrations from details specific to the demonstration scenario, such as exact object pose and geometry parameters. Without this separation, learning algorithms overfit to individual demonstrations and thus hinder generalization.

Our high-level solution to achieve this separation and generalization is to find the most appropriate inductive bias. In the field of machine learning, such inductive biases are a known precondition for generalization (Mitchell 1980). We will see in this section that there is a strong inductive bias that naturally enables the derivation of general policy from a single human demonstration, namely, environmental constraints. For clarity, we first present a high-level overview of the methodology that explains why this inductive bias enables generalization and identifies the fundamental components required to derive an EC-based policy. In the next section, we describe how these components are realized and implemented in a concrete LfD approach.

3.1 Environmental Constraints

An *environmental constraint* is a feature in the environment that can be used to identify the reference value of a behavior-generating controller (Eppner et al. 2015). It is important to note that the specific reference value is not specified a priori, but it is extracted from interactions with the environment during execution. Common examples include geometric features in the environment—surfaces, edges, kinematic joints, etc.—which can be used to simplify perception and control by exploiting information obtained through contact. Another set of environmental constraints provides visual rather than haptic references for behavior generation. Such constraints are leveraged in visual servoing. In both cases, the reference value is extracted from sensor data during execution.

This separation supports generalization because ECs arise from an object's articulation, which encodes its intended functionality. Objects designed for the same function often share the same EC structure; for example, doors, windows, scissors, and drawers exhibit recurring revolute or prismatic constraints. This invariance could be a consequence of human-centered design conventions that standardize how such objects can be used (Norman 2013). Leveraging this invariance, our approach represents a demonstration as a sequence of EC exploitations.

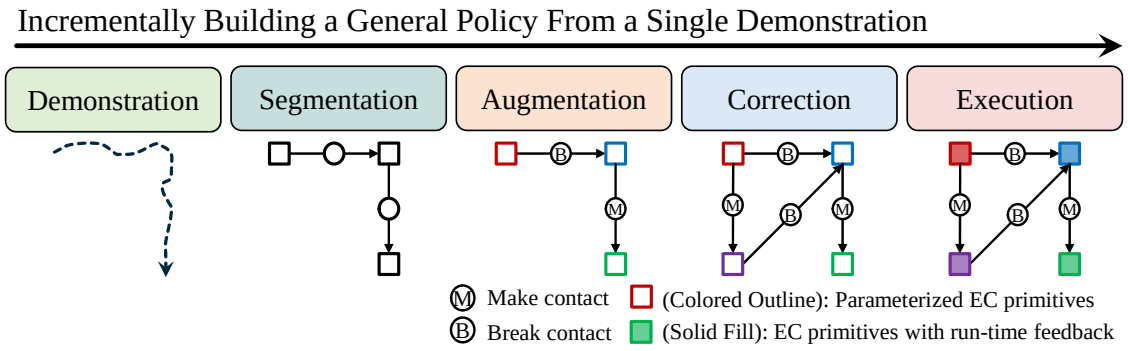


Figure 2. Overview of the four-stage LfD pipeline. A single human demonstration is segmented into a sequence of motion phases, where the exact demonstration actions that are specific to the demonstrated object instance are abstracted using the EC representation. In the next augmentation stage, the robot actively explores the environment around the demonstration to generate additional information about ECs, facilitating the identification of EC primitives (colored outlines) and contact transitions. When the robot encounters unforeseen task variations that are not captured by the EC representation, human corrections are added to expand the policy structure with new EC primitives (purple outline). Finally, during execution, EC primitives (solid fills) leverage sensory feedback to fill in instance-specific details, adapting to the particular instance at run time. Generalization is achieved by retaining the generalizable information needed for learning while leaving hard-to-model details to be completed through sensory feedback and corrections during execution.

Consider opening a drawer as an example. First, the robot visually servos toward the handle, whose visual features act as an EC that constrains the end-effector to a grasp pose. Second, after grasping, the robot simply pulls the drawer and lets the drawer's prismatic joint guide its motion. This motion stops when the drawer is fully opened, signaled by yet another EC, the stop of the drawer.

Therefore, ECs are the basic "currency" of a robot's actions in the real world. We will see in what follows that ECs represent the correct inductive bias for LfD. This inductive bias provides a suitable abstraction that avoids overfitting and enables systematic generalization.

3.2 Separating General From Specific

Building on ECs, we represent a policy as a hybrid automaton (Egerstedt 2000), consisting of ECEs (nodes) and transitions (edges). A node represents an ECE controller, implemented as a parameterized manipulation primitive. Each primitive characterizes a specific type of EC through admissible and constrained motion directions and generates motions to follow it accordingly. Nodes are connected via directed edges representing transition conditions. Transitions are triggered by the establishment of a new EC or the removal of a currently exploited one.

This policy structure cleanly separates *general* task information from *instance-specific* details. The *general* part consists of the EC types and their transitions, while the *instance-specific* information, such as the exact Cartesian trajectories or geometric parameters observed in a particular demonstration, is excluded in the policy representation. Instead, these details are recovered during execution, as described later.

Generalization emerges from this separation. For example, different door handles require different rotation angles before opening the door becomes possible, yet the underlying constraint-based behavior in the contact space—rotating until the door becomes pullable—remains the same. By leveraging this insight, our approach generates motion by following ECs rather than mimicking demonstrated trajectories for better generalization.

3.3 Completing the Environmental Constraint Using Augmentation

Since our policy is defined in contact space to enable generalization, deriving a good policy depends on correctly identifying ECs and their transitions from demonstrated trajectories. However, a single demonstration often lacks the required information to derive such a policy completely. In particular, when humans are familiar with a task, they tend to act exploitatively, accomplishing the task with minimal interaction with the environment (Li et al. 2023b). As a result, the recorded demonstration contains insufficient interaction (forces and moments) data for the robot to infer the underlying ECs. Moreover, distinct ECs can lead to similar motions (Subramani et al. 2018). Consequently, EC identification from demonstrations is often ambiguous.

To resolve this ambiguity, we extend LfD with self-guided exploratory actions. Rather than treating demonstrations as data to be directly imitated, we use them as guidance for autonomous, active interaction with the environment. Specifically, the robot replays a demonstrated trajectory while performing exploratory motions, generating additional constraint information and thereby completing the data required to derive a fully specified EC-based policy. We refer to this process as *augmentation*.

Using augmentation, the robot autonomously completes the partial EC description obtained in previous steps, enabling broader generalization, even for contact states that were not included in the demonstration data.

3.4 Corrective Demonstrations

While augmentation completes the demonstration data in terms of ECs, it does not change the overall structure of the policy. Naturally, a policy based on just one demonstration might fail when it meets unseen changes in task settings, e.g., appearances and local geometric differences, which are not captured by the EC representation. In such cases, corrective demonstrations are required.

Because EC-based behaviors are interpretable, it becomes easy for the policy to identify failures during execution and ask for corrective demonstrations. Moreover, because

the policy is compositional, each correction behavior can be incorporated as an additional EC primitive into existing policies, requiring targeted updates rather than extensive retraining. In this way, we incrementally extend policy generality to handle out-of-distribution task variations that are difficult to learn from limited data or to model explicitly.

3.5 Filling in Instance-Specific Details to Complete the Policy

After constructing a generalized EC-based policy through augmentation and corrective demonstrations, the instance-specific details previously abstracted away to facilitate learning—such as exact object poses, contact locations, and local geometries—must be recovered to fully parameterize the policy. This happens during execution of the policy based on sensory feedback obtained from interaction with the current task instance.

To this end, we leverage compliant control, which allows the robot to automatically adapt its motion to the encountered ECs. The resulting force and motion feedback is used to update the internal EC model online. This online adaptation shifts the burden of handling instance-specific variations from learning to compliant control, enabling in-context adaptation that supports generalization.

Taken together, the separation of general and instance-specific information, combined with augmentation, corrective demonstrations, and compliant adaptation, forms a complete pipeline for LfD (see Fig. 2). As we will demonstrate in the following sections, simple and classical components composed according to this pipeline are already sufficient to build an interactive LfD approach that achieves substantial generalization in complex, multi-stage contact-rich manipulation tasks.

4 Implementing LfD for Multi-Stage Contact-Rich Manipulation

We now describe how the proposed LfD pipeline is implemented to teach manipulation tasks that combine in-contact motions (e.g., insertion and threading) and free-space motions (e.g., grasping and placing) to a robot.

A key observation is that the interaction patterns within these tasks can be captured by four types of ECs: *translation*, *rotation*, *plane*, and *free-space*. Translation and rotation reflect prismatic and revolute joints common in articulated objects, while plane constraints frequently arise in assembly tasks where objects must slide along surfaces to achieve alignment or insertion. The free-space primitive uses visual feedback to reproduce contact-free motions, such as grasping or placing objects. We model these transitions between EC primitives as discrete events, including *contact-making*, *contact-breaking*, and *gripper* events (i.e., grasp and release).

In the remainder of this section, we first describe demonstration recording and segmentation, then introduce the EC primitives that form the policy. We next present augmentation strategies that support policy construction and corrective demonstrations for policy refinement. Finally, we describe how sensing and compliant control are used during execution to recover the instance-specific details.

4.1 Demonstration and Segmentation

We collect a kinesthetic demonstration using a Franka Emika Panda robot (Haddadin et al. 2022) with a parallel gripper, a wrist-mounted force/torque sensor, and an end-effector–mounted RGB-D camera. Each demonstration waypoint includes the end-effector pose $\mathbf{T}_{\mathrm{BE}}$, where B denotes the base frame and E the end-effector frame, the gripper state $g \in (0,1)$, the measured wrench $F \in \mathbb{R}^6$, and the RGB-D image I.

We segment a demonstration based on multi-modal sensory signals. We first use gripper state changes to segment grasp and release actions. Since gripper states alone do not fully capture constraint transitions, we further analyze force measurements. We compute the force norm and apply Total Variation Denoising (TVD) (Selesnick 2017) to smooth the signal while preserving edge information. We then identify rising and falling edges using a 10 N threshold as segments (Eiband et al. 2019). Segments with an average force below 2 N are labeled as free-space motions, while others are labeled as in-contact motions.

Since individual in-contact segments may encompass multiple ECs, we further refine these segments using Zero-Velocity-Crossing (ZVC) (Fod et al. 2002). This method identifies discontinuities in end-effector velocity that typically correspond to transitions between ECs. We avoid over-segmentation by removing any segments where the robot moves less than 2 cm.

After segmenting the demonstration, we have a sequence of motion phases. Each phase will be turned into one of our four EC primitives (Fig. 4). To keep the explanation clear, we will first describe how these EC primitives work and then explain how the system identifies which one to use for each part of the task.

4.2 EC Primitives

Free-Space Primitive. This primitive is designed to replicate contact-free motions like grasping an object. We implement this primitive based on Trajectory Transfer (Vitiello et al. 2023), where the robot aligns its current visual input with a reference image from the demonstration via SIFT-based visual servoing (Chaumette et al. 2016). The visual alignment places the end-effector in the same pose relative to the object as in the demonstration. This allows the subsequent trajectory and EC primitives to execute in an object-centric frame, thereby enabling generalization across different object poses.

Methods based on Trajectory Transfer typically utilize the first frame of the demonstration as the visual reference. However, in multi-stage tasks, these methods might fail after a grasp because the wrist-mounted camera is too close to the object to extract enough features for subsequent visual alignment. To mitigate this, we propose a selection strategy that identifies the reference image at the waypoint of maximum elevation in the task space (i.e., the peak z-coordinate relative to the initial pose). Our revised procedure involves: (1) replaying the trajectory up to that waypoint, (2) aligning the current view to the reference image, (3) transforming and executing the rest of the trajectory in the aligned end-effector pose, and (4) executing the gripper actions according to the demonstrated gripper states. This

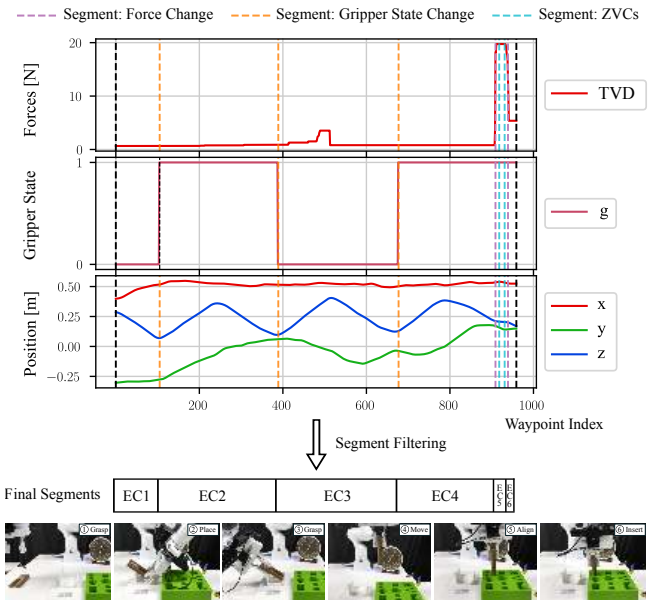


Figure 3. Segmentation result of a demonstration for a single-object insertion task from Functional Manipulation Benchmark (Luo et al. 2025). The top plots show TVD force signals, gripper states, and end-effector positions, indexed by waypoint of a demonstration. Vertical dashed lines indicate segmentation from force thresholds (purple), gripper state changes (orange), and ZVCs (cyan). We first merge these segments and then remove those whose total displacement is less than 2 cm. The bottom row shows six final segments and snapshots of the robot behavior during each motion phase.

design allows the free-space primitive to handle repeated grasping and repositioning (see Fig. 4).

Taken together, the free-space primitive is parameterized by a reference image, an end-effector trajectory, and gripper states. A SIFT-based visual servoing controller is used to minimize feature errors between current and reference views (Costanzo et al. 2022).

Plane Primitive. The plane primitive reproduces motions along flat surfaces, such as sliding or guiding objects into slots. To ensure the policy remains independent of absolute Cartesian coordinates, we fit a planar model to the segmented demonstration trajectory to extract the surface normal vector $\hat{\mathrm{n}} \in \mathbb{R}^3$. During execution, the specific plane location is instantiated by the end-effector pose at the moment of making contact with the surface. The primitive then transfers the demonstrated trajectory to the local end-effector frame and replays it while maintaining contact through the Cartesian compliant controller (Sec. 4.6).

After replaying the trajectory, the robot performs SIFT-based visual servoing toward the reference image from the last waypoint of the motion phase and stops when a displacement threshold is reached. It then executes a spiral search on the surface (Chhatpar and Branicky 2001). These two steps help generalize to uncertain or shifted hole locations.

A plane primitive is parameterized by an end-effector trajectory and a reference image from the segmented motion phase. It terminates upon detecting: a tangential force exceeding a threshold (making-contact event), a displacement along the surface normal direction exceeding a threshold (breaking-contact event), or no contact is found after finishing the spiral search (non-contact event).

Translation Primitive. The translation primitive models 1-DOF linear motions, such as pulling a drawer or inserting an object into a hole. Like the plane primitive, the translation primitive avoids using a fixed prismatic model from the demonstration, but maintains a prismatic joint tracker that online estimates the admissible motion direction based on observed end-effector displacements. As evidenced in our real-world experiments, this online adaptation allows the robot to generalize to unseen object geometries or inaccurate models.

The prismatic joint tracker is implemented as a particle filter (Li and Brock 2022), where each particle represents a motion direction $\hat{\mathrm{u}}^{\mathrm{t}} \in \mathbb{R}^3$ parameterized in spherical coordinates $(\theta^{\mathrm{t}}, \phi^{\mathrm{t}})$ (the superscript t denoting the translational motion type). During the prediction step, particles are perturbed with zero-mean Gaussian noise. We then sample a candidate direction from the particle distribution and execute a compliant motion along it. If the resulting linear motion exceeds 3 mm, the filter increases the weights of particles that are close to the observed motion direction; otherwise, it skips the update, which increases the distribution's variance and thus encourages more exploration in subsequent steps.

The translation primitive is initialized by fitting a line to the segmented position trajectory to obtain $\hat{\mathrm{u}}^{\mathrm{t}}$. Execution terminates upon detecting a contact change, either when no update step is performed after a maximum number of sampling attempts (making-contact event), or when the robot detects significant displacement in a previously constrained direction (breaking-contact event).

Rotation Primitive. The rotation primitive reproduces motions around revolute constraints, such as rotating a handle or screwing a cap. The rotation primitive follows the same sample-execute-update process as the translation primitive: it maintains a revolute joint tracker, samples a desired end-effector pose from the current estimate of the revolute joint parameters, executes it with compliant control, and updates the estimate based on the observed motion. This process allows the robot to adapt to local variations in the revolute constraint parameters and thus generalize across unseen scenarios.

The revolute joint tracker is implemented as the Extended Kalman Filter (EKF) presented in Martín-Martín and Brock (2022), which maintains a belief over the screw parameters $(\hat{\mathrm{u}}^{\mathrm{r}}, \mathrm{p}^{\mathrm{r}}, q^{\mathrm{r}}, \dot{q}^{\mathrm{r}})$, where $\hat{\mathrm{u}}^{\mathrm{r}} \in \mathbb{R}^3$ is the rotation axis (with the superscript r denoting the rotational motion type), $\mathrm{p}^{\mathrm{r}} \in \mathbb{R}^3$ is a point on the axis, $q^{\mathrm{r}} \in \mathbb{R}$ is the rotation amount, and $\dot{q}^{\mathrm{r}} \in \mathbb{R}$ is its angular velocity.

At each step, the EKF propagates q using $\dot{q}^{\mathrm{r}}$ and injects process noise into the state. A rotation axis is then sampled from the EKF belief to compute the next motion command $z \in se(3)$ using the measurement model:

$$z = \begin{pmatrix} q^{\mathrm{r}}\,\hat{\mathrm{u}}^{\mathrm{r}} \\ -q^{\mathrm{r}}\,\hat{\mathrm{u}}^{\mathrm{r}} \times \mathrm{p}^{\mathrm{r}} \end{pmatrix},$$

where the first term corresponds to the angular motion and the second term to the linear motion. The desired end-effector pose is then computed as $\mathbf{T}_{\mathrm{BE}} = \exp(z^{\wedge})\ \mathbf{T}_{\mathrm{BE}_{\mathrm{init}}}$, where $\exp(\cdot^{\wedge})$ denotes the matrix exponential map and $\mathbf{T}_{\mathrm{BE}_{\mathrm{init}}}$ is the initial end-effector pose at the start of the primitive. Similar to the prismatic joint tracker, the EKF

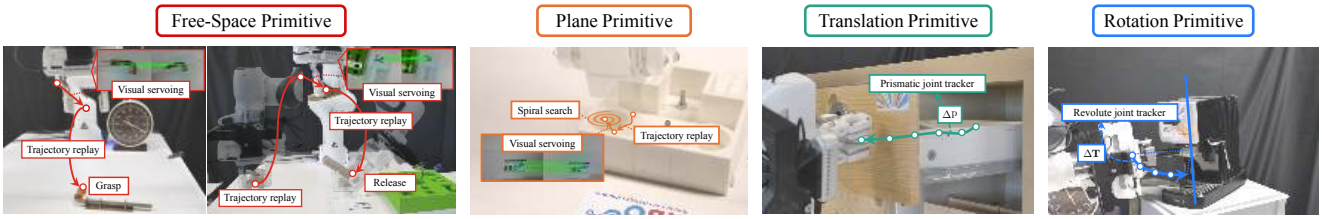


Figure 4. Overview of the four EC primitives. The **free-space primitive** aligns the end-effector to the target object through visual servoing using a wrist-mounted camera, enabling adaptation to changes in object pose. It then replays the transferred demonstrated trajectory to reproduce contact-free motions. The **plane primitive** first replays the transferred demonstration trajectory along a surface and then refines its position through visual servoing and spiral search, enabling generalization to uncertain or shifted hole locations. The **translation and rotation primitives** generate compliant motions that follow prismatic and revolute constraints, respectively, and use observed end-effector movements to update their internal models, allowing generalization to different object geometries. Together, these four EC primitives generate motions that generalize across unseen *object poses* and *geometries*.

only performs an update when the observed linear motion exceeds 1 cm or the angular motion exceeds 2°; otherwise, the update is skipped. Skipping the update increases the uncertainty of the states and encourages the robot to explore different rotation axes in the subsequent sampling steps. This prevents the robot from getting stuck due to inaccurate model parameters.

The rotation primitive is initialized using the screw parameters $\hat{\mathrm{u}}^{\mathrm{r}}$ and p^{r} obtained by fitting a revolute constraint model to the segmented demonstration trajectory. The termination conditions are the same as for the translation primitive.

4.3 Augmentation

After segmentation, we must instantiate an EC primitive for each motion phase by assigning it to the correct EC type. This assignment is non-trivial for in-contact motion phases due to inherent kinematic ambiguities. For example, an end-effector trajectory following a prismatic or revolute joint may also satisfy a plane constraint that happens to be a line or circle. It is also difficult to determine whether a primitive terminates and transits because of making a new contact or breaking an existing contact because noisy force measurements can obscure contact-changing events.

To address these challenges, we introduce a *contact-based augmentation* strategy in which the robot replays the demonstration trajectory while performing exploratory probing actions to actively verify the candidate constraint directions and contact transitions. Additionally, we propose a *vision-based augmentation* strategy to improve visual alignment in free-space primitives by enhancing image features.

4.3.1 Contact-Based Augmentation As described earlier, we need to assign an EC type to a segmented motion phase. To do so, we first classify motion phases as contact-free or in-contact using force thresholds. A free-space primitive is directly assigned to contact-free motions. For in-contact motions, we initially distinguish translation and rotation ECs using an angular displacement threshold of 30°. To resolve remaining geometric ambiguities, we replay the demonstration and control the robot to probe a direction $\hat{\mathrm{d}} \in \mathbb{R}^3$ at the midpoint of each in-contact motion:

$$\hat{\mathrm{d}} = \begin{cases} \hat{\mathrm{u}}^{\mathrm{t}} \times \hat{\mathrm{n}}, \text{ translation} \\ (\mathbf{I} - \hat{\mathrm{u}}^{\mathrm{r}}\,\hat{\mathrm{u}}^{\mathrm{r,T}})\,(\mathrm{p} - \mathrm{p}^{\mathrm{r}}), \text{ rotation} \\ \hat{\mathrm{n}}, \text{ plane} \end{cases}$$

where $\mathrm{p} \in \mathbb{R}^3$ is the end-effector position, $\hat{\mathrm{n}}$ is the surface normal direction, $\hat{\mathrm{u}}^{\mathrm{t}}$ and $\hat{\mathrm{u}}^{\mathrm{r}}$ are the translation and rotation axes, and $\mathrm{p}^{\mathrm{r}} \in \mathbb{R}^3$ is a point of the rotation axis. These parameters are computed for each motion phase by fitting geometric constraint models to the segmented trajectory using the method of Subramani et al. (2018).

If the robot observes a constraint along the direction $(\mathbf{I} - \hat{\mathrm{u}}^{\mathrm{r}}\,\hat{\mathrm{u}}^{\mathrm{r,T}})\,(\mathrm{p} - \mathrm{p}^{\mathrm{r}})$, the motion phase is classified as a rotation EC, since this direction would remain unconstrained for a plane EC. Similarly, if no constraint is detected along $\hat{\mathrm{u}}^{\mathrm{t}} \times \hat{\mathrm{n}}$, the motion phase corresponds to a plane EC, as this direction is constrained by a translation EC.

Following our prior work (Li et al. 2023b), we further verify contact-changing events between consecutive motion phases. After disambiguating the EC type of a motion phase, the robot compliantly executes the trajectory of the next motion phase. If the motion is blocked by a constraint, we classify the transition as a breaking-contact event; otherwise, we label it as a making-contact event. Examples of this contact-based augmentation strategy are shown in Extension 1 and Extension 4.

Overall, the contact-based augmentation strategy actively verifies constraint directions around the demonstration trajectory, generating additional contact information that supports the recognition of EC types and contact transitions.

4.3.2 Vision-Based Augmentation The performance of visual alignment within the free-space primitive depends heavily on the viewpoint invariance of SIFT features. To improve alignment performance, we introduce a vision-based augmentation that prunes SIFT features that are prone to cause mismatches under changing views.

Given a free-space primitive, we first reset the object to its initial demonstration pose. The robot then selects a reference image, extracts SIFT features, and projects them into 3D using the depth data. Next, it executes a scripted motion that actively varies the camera viewpoint. For each newly captured image, we match the observed SIFT features to the reference features using FLANN-based matching (Bradski 2000) and project the matched keypoints into 3D.

Since the target object remains static, the quality of each feature match can be evaluated by measuring the 3D

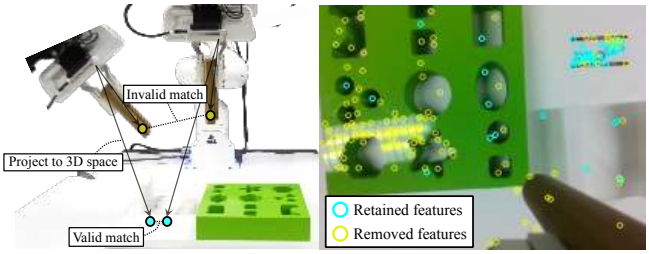


Figure 5. Vision-based augmentation to filter unstable features. By actively changing the robot's viewpoint and evaluating feature matching performance (left), the vision-based augmentation strategy filters out features that are not consistent under viewpoint changes (right). This includes features caused by light reflections on the insertion board surface, which are prone to mismatching and sensitive to lighting variation. It also filters out features on the inserted object. Because these features move with the robot, they are inconsistent with changing viewpoints and would introduce errors to the visual alignment process.

Euclidean distance between the matched feature and its original position in the reference image. Matches with a displacement smaller than 2 cm are considered valid. For each reference SIFT feature, a matching score is computed as the normalized count of valid matches observed across all viewpoints. Features with scores below 0.2 are labeled as viewpoint-inconsistent and removed.

As exemplified in Fig. 5 and Extension 14, the proposed vision-based augmentation strategy effectively filters out features prone to mismatching. The results presented in Appendix A further demonstrate that the enhanced visual alignment improves the generalization to lighting changes.

4.4 Policy Refinement With Human Corrections

Following augmentation, the resulting policy comprises a sequence of EC primitives linked by discrete contact-state transitions and gripper events. Although this policy captures the general task structure, executions may still fail when the policy encounters out-of-distribution variations that are not modeled by the EC representation. However, the discrete nature of the policy inherently facilitates failure detection. By continuously monitoring transition events, the policy can evaluate whether an EC primitive terminates nominally. For instance, a missed insertion manifests as the absence of an expected breaking-contact event, whereas a lost grasp yields an unexpected gripper state.

Once an execution failure is detected, the user can provide a corrective demonstration. Specifically, we implement two ways to incorporate human corrections. If a free-space primitive ends incorrectly (e.g., a lost grasp), the user provides a new demonstration that replaces the reference image and associated trajectory, allowing the primitive to adapt to the new scenario, such as changes in object appearance. For a contact-based primitive, the user demonstrates a recovery trajectory that brings the robot to a state where the original failed EC can resume execution. A new primitive of the same EC type is instantiated and linked to the failed EC through the newly observed contact transition. When this corrective primitive is triggered, the robot first executes the recovery trajectory and then continues with the associated EC-based motions. An example of the correction process is shown in Extension 6.

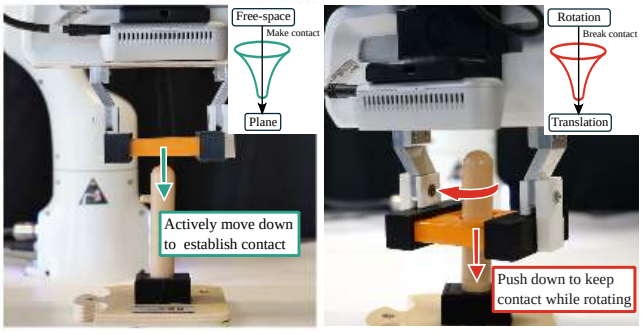


Figure 6. Illustration of our funneling strategies in the puzzle insertion task. **Left**: After a free-space primitive, the robot stops just short of contact. Transitioning to the plane primitive from this state would erroneously trigger a breaking-contact event, leading to premature state transition and subsequent manipulation failure. Our funneling strategy addresses this by moving along the plane's constraint direction to establish contact, compensating for positional uncertainties. **Right**: During a rotation primitive, the robot applies a force along the constrained axis (the admissible direction of the subsequent translation primitive) to maintain contact. This simplifies the detection of the desired contact-breaking event.

4.5 Sequencing EC Primitives as Funnels

During the execution of an EC-based policy, we enhance robust transitions between EC primitives by intentionally establishing or maintaining contact with the environment. This is grounded in the sequential composition of funnels (Burridge et al. 1999), where each EC primitive acts as a funnel that guides the system into the specific contact state required by the subsequent primitive.

Specifically, we implement two funneling strategies (see Fig. 6). First, when transitioning from a free-space motion to an in-contact primitive, the policy proactively commands a motion along the anticipated constraint direction of the upcoming EC to establish contact. Second, when a breaking-contact transition is expected, the robot deliberately maintains contact by applying force along the admissible direction of the next EC. These two strategies deliberately establish or maintain contact to expose the desired contact feedback during execution, facilitating transitions between EC primitives and improving robustness to uncertainties in pose or geometry.

4.6 Cartesian Compliant Controller

An EC-based policy generates behaviors that alternate between free-space and in-contact motions. Free-space motions benefit from high-stiffness control to ensure precise execution, whereas in-contact motions require more compliance to adapt to local geometry. Also, the robot must handle frequent and unpredictable contact transitions. Thus, we need a compliant controller that is adaptive and contact-aware.

We design a Cartesian compliant controller that meets these requirements, building on the equilibrium-point compliance formulation of Jain and Kemp (2009). The key idea is to integrate wrench feedback from a wrist-mounted force/torque sensor into the trajectory generator that modulates the desired equilibrium pose.

Given a target pose G in the base frame B, namely $\mathbf{T}_{\mathrm{BG}}$, we first interpolate the goal in exponential coordinate $\Delta\psi_{\mathrm{t}} \in se(3)$ subject to an element-wise twist limit $V_{\mathrm{m}} \in \mathbb{R}^6$ by

$$\Delta\psi_t = \max\left(\min\left(\log\left((\mathbf{T}^*_{\mathrm{BE}_{\mathrm{t}-1}})^{-1}\,\mathbf{T}_{\mathrm{BG}}\right)^{\vee}, V_{\mathrm{m}}\,\Delta t\right),\ -V_{\mathrm{m}}\,\Delta t\right) \tag{1}$$

where $\Delta t = 0.001$ corresponds to a 1kHz control frequency, $\mathbf{T}^*_{\mathrm{BE}_{\mathrm{t}-1}}$ is the desired pose at the last time step $\mathrm{t}-1$, and $\log(\cdot)^{\vee}$ denotes the logarithmic map. We further constrain $\Delta\psi_t \in se(3)$ based on the difference between the element-wise absolute value of the observed wrench $|F_{\mathrm{obs,t}}| \in \mathbb{R}^6$ and the predefined wrench limit $F_{\mathrm{m}} \in \mathbb{R}^6$:

$$\Delta F_{\mathrm{t}} = F_{\mathrm{m}} - |F_{\mathrm{obs,t}}| \tag{2}$$

$$\Delta\psi^*_{\mathrm{t}} = \begin{cases} \tanh(|\Delta F_{\mathrm{t}}|)\,\Delta\psi_{\mathrm{t}}, & \Delta F_t > 0 \\ \mathrm{sign}\,(F_{\mathrm{obs,t}})\,\tanh(|\Delta F_{\mathrm{t}}|)\,V_{\mathrm{m}}\,\Delta t, & \text{otherwise} \end{cases} \tag{3}$$

where the $\tanh(\cdot)$ function acts as a smooth factor. When $\Delta F_{\mathrm{t}} < 0$, i.e., when the force limits are exceeded, $\mathrm{sign}(\cdot)$ flips the interpolation direction to reduce the observed wrench $F_{\mathrm{obs,t}}$. After obtaining the wrench-limited relative pose $\Delta\psi^*_{\mathrm{t}} \in se(3)$, we compute the desired end-effector pose by $\mathbf{T}^*_{\mathrm{BE}_{\mathrm{t}}} = \mathbf{T}^*_{\mathrm{BE}_{\mathrm{t}-1}}\,\exp(\Delta\psi^{*\wedge}_{\mathrm{t}})$. We then compute the desired motor torques $\tau_{\mathrm{cmd}} \in \mathbb{R}^7$ following the operational space formulation (Khatib 1987)

$$\tau_{\mathrm{cmd}} = J^T\ \left(K_{\mathrm{p}}\,e_{\mathrm{pose}} - K_{\mathrm{d}}\,V_{\mathrm{obs}}\right) + \tau_{\mathrm{dyn}}, \tag{4}$$

where $K_{\mathrm{p}} \in \mathbb{R}^{6\times6}$ and $K_{\mathrm{d}} \in \mathbb{R}^{6\times6}$ are positive symmetric controller gain matrices for stiffness and damping respectively, $e_{\mathrm{pose}} \in \mathbb{R}^6$ is the error between the observed and desired end-effector pose, $V_{\mathrm{obs}} \in \mathbb{R}^6$ is the observed end-effector twist, $J^T \in \mathbb{R}^{7\times6}$ is the Jacobian transpose, $\tau_{\mathrm{dyn}} \in \mathbb{R}^7$ represents the torques needs to be compensated such as gravity and Coriolis force.

We evaluate the controller using a door-lock manipulation task requiring the robot to establish contact with the environment. As illustrated in Fig. 7, our controller successfully maintains a desired contact force even with high stiffness. Consequently, this controller allows us to directly specify the desired interaction force simply by setting the appropriate wrench limits. This mitigates impact forces during contact transitions, reduces the risk of damaging the manipulated object, and avoids tedious tuning of control gains. These features are essential for implementing EC exploitation behaviors on real robots.

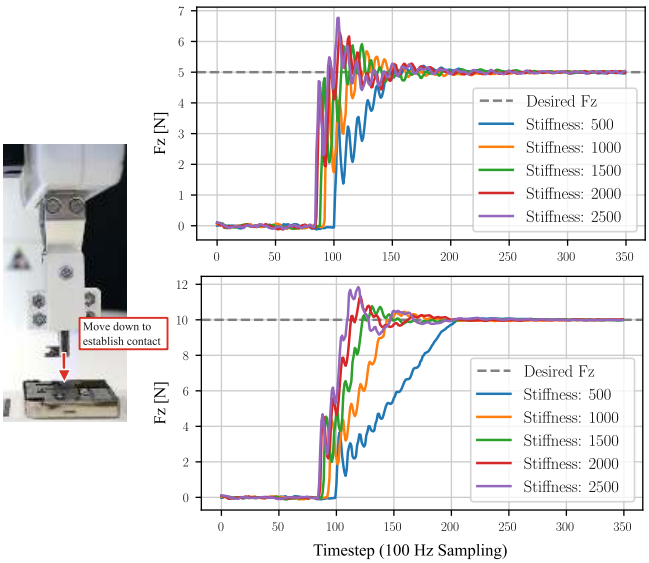


Figure 7. Observed z-axis forces as the robot approaches the lock surface at 2 cm/s. The controller is evaluated with predefined force limits of 5 N and 10 N across stiffness settings from 500 N/m to 2500 N/m. The force profiles show that the controller reliably maintains the target contact force under various stiffness settings, enabling safe interaction and simplifying parameter tuning.

5 Experiments

We evaluate the generalization capability of our approach on seven real-world multi-stage contact-rich manipulation tasks with rigid objects, spanning insertion, threaded fastening, articulated-mechanism manipulation, lock operation with a key, and long-horizon assembly (see Fig. 1). These tasks are challenging because they require precise physical interaction and multi-stage sequential execution. Our extensive real-world experiments demonstrate that ECs are the appropriate inductive bias for generalization by (i) abstracting task-general structure based on ECs from instance-specific Cartesian trajectories, (ii) enabling online adaptation to uncertain or imprecise constraint models, and (iii) supporting failure detection and targeted policy refinement through human corrections during execution.

Below, we describe these tasks and their challenges, followed by an outline of the assumptions used in the experiments.

5.1 Task Overview and Challenges

Insertion Puzzle. The robot inserts a grasped part into a puzzle board and then performs a sequence of actions so that the part passes through internal pins. While these puzzles share similar visual appearances, each requires a distinct Cartesian trajectory to complete. Another challenge lies in combining precise insertion with large rotations (up to 270°).

Door Locks with Keys. The robot inserts a key into a lock and rotates it by 120° to change the lock's state. Compared to insertion puzzles, this task is more sensitive to pose uncertainty due to tighter tolerances.

Coffee Machine. The robot inserts a portafilter and rotates it by approximately 45° to tighten it. This task requires the robot to operate near its joint limits, which reduces kinematic manipulability and increases execution noise. Additionally, we challenge the system with imprecise EC models.

Latches and Drawers. These tasks require the robot to grasp and operate mechanisms with multiple DOFs. The key challenge is that critical task parameters for deriving Cartesian trajectories, such as the rotation angles of latches and handles, are difficult to infer from visual observations. Even when such trajectories are available, executing them is prone to failure due to unmodeled slippage between the robot's gripper and the manipulated object during execution.

Functional Manipulation Benchmark and FurnitureBench. These tasks are taken from two recent long-horizon contact-rich manipulation benchmarks and require sequential actions, including grasping, placing, regrasping,

inserting, and screwing. Prior work has shown that learning such behaviors end-to-end remains challenging even with large numbers of demonstrations, due to the long-horizon structure and fine-grained contact interactions.

Overall, these tasks involve complex contact dynamics, partial observability of task parameters, variations in object geometry, imprecise geometric parameters inferred from a demonstration, and long-horizon execution under full 6D pose uncertainty. In the following experiments, we demonstrate how using ECs as the inductive bias enables policies to effectively address these challenges.

5.2 Experimental Assumptions

To ensure a fair interpretation of the experimental results, we explicitly state the key assumptions and revisit them in Sec. 6, which also outlines promising directions for relaxing these assumptions.

- We assume objects have sufficient and transferable SIFT features for the free-space primitive. We achieve this by using textured image stickers.
- In the insertion puzzle, door lock, and coffee machine tasks, we assume the robot begins with a firm grasp of the manipulated object.
- We assume that the provided demonstrations are segmented into appropriate motion phases without mis-segmentation.
- We assume that objects are placed within the robot's reachable workspace such that the task can be executed without violating joint limits.
- We assume task-relevant fixtures, such as insertion boards and locks, are rigidly mounted on the environment.
- We assume task-specific control parameters and contact-transition thresholds can be set a priori (see Table 1); these constitute minor tuning and do not change the LfD pipeline.

5.3 Insertion Puzzles

We first test our method on insertion puzzle tasks (Fig. 8) to evaluate generalization across object geometries. We record a demonstration using the orange puzzle. Our approach automatically segments this demonstration into seven motion phases and then performs augmentation to disambiguate ECs and contact-changing events. The resulting policy consists of seven EC primitives. The policy construction process is shown in Extension 1. Table 1 lists the control parameters and the thresholds used for detecting contact-changing events.

We compare our method to eight baselines with different policy representations. BC (Young et al. 2021) and VINN (Pari et al. 2022) take wrist camera images as input and output delta end-effector poses. ACT (Zhao et al. 2023) uses images from a wrist camera and a static camera along with proprioception to predict joint position trajectories. DP (Chi et al. 2025) and FM (Zhang and Gienger 2024) share the same input modality as ACT but predict desired end-effector poses, which are then tracked by our Cartesian compliant controller. DP-Force and FM-Force extend DP and FM with an additional MLP head that processes the wrist-mounted force/torque signal, following the architecture proposed in (Luo et al. 2025). Finally, Trajectory Transfer (Vitiello et al. 2023) is included as a classical LfD baseline that uses a single demonstration without our augmentation phase. We record 30 demonstrations for the end-to-end learning methods and 1 demonstration for Trajectory Transfer.

We run 10 trials per method and report success rates. The puzzle board is rigidly fixed on the table. Each trial

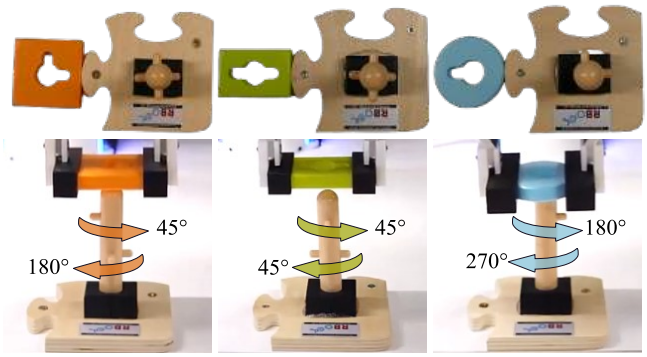


Figure 8. The three insertion puzzles used in our experiments. While each puzzle requires a distinct Cartesian trajectory, their motions in the contact space remain the same.

Table 1. Control and event thresholds used in the experiments.

Param.	Description	Value
$K_{p,trans}$	stiffness gains for the translation primitive [N/m, Nm/rad]	diag[500, 500, 500, 0, 0, 0]
K_p	stiffness gains for other primitives [N/m, Nm/rad]	diag[1000, 1000, 1000, 200, 200, 200]
K_d	damping gains [Ns/m, Nms/rad]	diag[10, 10, 10, 5, 5, 5]
V_m	twist limit [m/s, rad/s]	[0.03, 0.03, 0.03, 0.3, 0.3, 0.3]
F_m	wrench limit [N, Nm]	[5, 5, 5, 2, 2, 2]
r	radius of spiral search trajectory	3 cm
MCE^p	force threshold that triggers making-contact event for a plane primitive	5 N
MCE^t and MCE^r	maximum sampling attempts for a making-contact event in translation and rotation primitives	30
BCE	displacement threshold for a breaking-contact event	0.8 cm

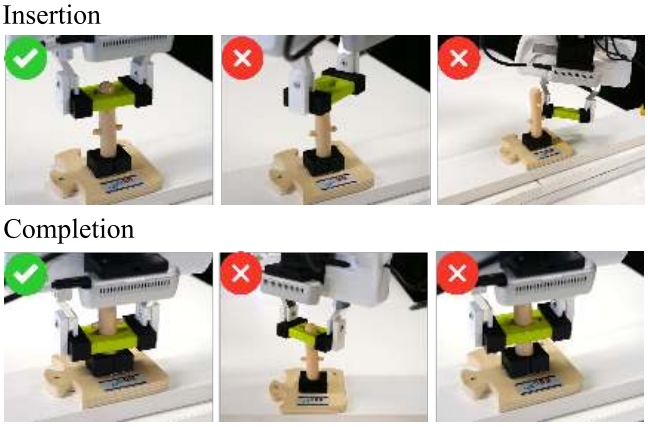


Figure 9. Illustration of success and failure modes in the puzzle insertion task, divided into two stages: *Insertion* (top) and *Completion* (bottom). A trial is considered a successful *insertion* if the part is correctly placed into the puzzle board. Failure modes in this stage include misalignment or prematurely lowering the part. In the *completion* stage, success requires guiding the part through the internal pins via two sequential and precise rotations. Failures at this stage include incorrect rotation due to perceptual aliasing and incomplete insertion resulting from perceptual and control inaccuracies.

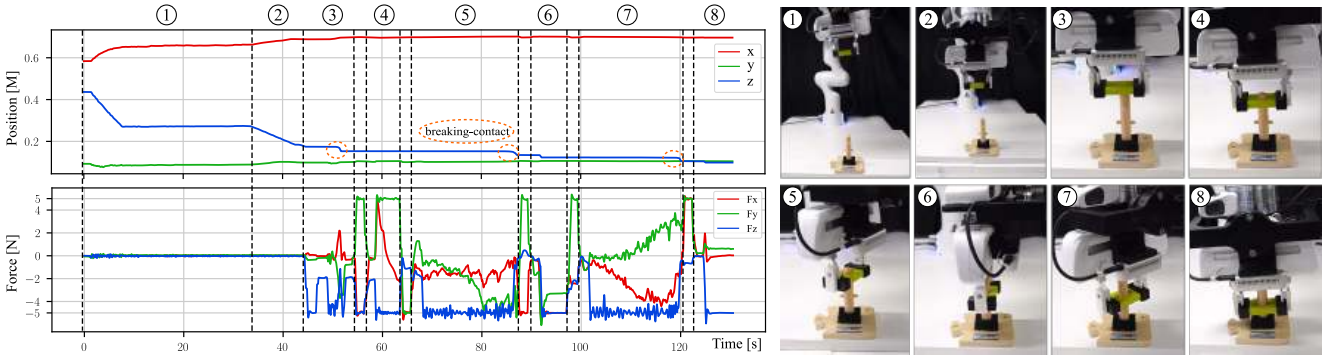


Figure 10. Solving a green puzzle by following a sequence of ECs. ① The robot first visually aligns the end-effector to the puzzle and transfers the demonstration trajectory to the new pose. ② It replays the trajectory and purposefully pushes down until making contact with the puzzle. ③ It then executes the plane primitive that replays the transferred trajectory on the plane, followed by visual servoing and a spiral search action to reduce positional uncertainties, terminating upon successful insertion (a breaking-contact event). ④ The robot then pushes the part down by following the prismatic constraint until it makes contact with the pins of the insertion board. ⑤ It rotates the part while pushing, breaking contact again to terminate the rotation primitive. ⑥-⑦-⑧ The robot repeats the translation and rotation motions until the grasped part is completely fitted into the puzzle. The force plot shows that the robot generates sequential compliant motions sequenced by making- or breaking-contact events, rather than following absolute positional trajectories, which leads to generalization. It also shows that our compliant controller is able to effectively regulate the force within the predefined threshold (5 N; see Table 1) during the manipulation, ensuring safe forceful interaction.

is initialized from a nominal end-effector pose, subject to uniformly sampled rotational perturbations of $U(-15°, 15°)$ across all three axes. This requires the robot to generate actions in full 6D space. A trial counts as a successful *insertion* if the part is inserted on the top of the board, and a successful *completion* if it passes through the lowest pins (see Fig. 9).

Table 2 summarizes the results. BC and VINN perform worst, failing to finish the insertion task in all trials. As observed, BC often rotates the end-effector before contacting the board, leading to out-of-distribution visual observations and causing compounding errors. VINN reduces compounding errors by retrieving actions from demonstrations, but it fails due to poor action transitions—it consistently executes the push-down action too early during the approach, leading to failure.

By contrast, we observe that ACT and generative-model-based methods perform well during the initial insertion stage on the trained orange puzzle, with ACT, DP, and FM achieving 10, 10, and 8 successful trials, respectively. This performance gain is attributed to the richer sensory input (images from both wrist and static cameras plus proprioception) and the prediction of sequences of absolute end-effector poses, which are more stable than delta movements. Furthermore, DP and FM could partially transfer the insertion skill to the blue puzzle (achieving 8 and 6 successful trials, respectively), while all methods struggle with the green puzzle. We attribute this to changes in the size and color of the inserted parts, which shift the contact positions and also change the observations from both cameras, thereby degrading policy performance. Additionally, including force inputs yields no notable performance improvement for DP-Force and FM-Force.

Trajectory Transfer completes the task on the orange puzzle in 9 of 10 trials. Furthermore, it demonstrates partial generalization to the green and blue puzzles, achieving insertion-stage success in 7 and 8 trials, respectively. However, because this method relies on tracking spatially adapted Cartesian trajectories, it inherently fails to accommodate puzzles with differing internal geometries during the completion stage. Additionally, it remains highly sensitive to visual alignment errors, which is shown in the Appendix A.

Despite varying degrees of success in the *insertion* stage, none of the baseline methods were able to derive a policy that generalizes from the orange puzzle to other puzzles.

In contrast, our approach successfully accomplishes the task on the orange puzzle in all 10 trials and directly generalizes to the green and blue puzzles, as shown in Extension 2. This strong generalization stems from generating motions with respect to ECs in the contact space rather than mimicking exact demonstration actions. For example, our policy rotates the grasped part until it passes through a pin, regardless of the specific rotation angles performed (see Fig. 10). As a result, a single demonstration is sufficient to reliably accomplish the task and generalize across geometric variations.

The efficient generalization of our approach highlights that ECs serve as a suitable abstraction, preserving the general task structure while discarding instance-specific details (e.g., absolute Cartesian trajectories), thereby enabling a policy learned on a single instance to directly generalize to novel geometries.

While these end-to-end methods underperform on this task, they are designed for broader task domains and are typically evaluated in settings with larger datasets. This performance difference should therefore not be interpreted as evidence against end-to-end methods in general. Rather, it indicates that for narrower task regimes such as multi-stage contact-rich manipulation, the right inductive bias can provide a substantial advantage in sample-efficient generalization. We return to this no-free-lunch tradeoff in Sec. 7.

5.4 Door Locks with Keys

We now turn to a more challenging task: door lock manipulation. Fig. 11 illustrates two locks used in the experiment. We first record a demonstration on Lock 1. Our approach segments this demonstration into three motion phases, augments them through exploratory actions to

Method	Orange		Green		Blue	
	Insertion	Completion	Insertion	Completion	Insertion	Completion
BC	0/10	0/10	0/10	0/10	0/10	0/10
VINN	0/10	0/10	0/10	0/10	0/10	0/10
ACT	10/10	0/10	0/10	0/10	1/10	0/10
DP	10/10	0/10	1/10	0/10	8/10	0/10
FM	8/10	0/10	0/10	0/10	6/10	0/10
DP-Force	7/10	0/10	1/10	0/10	8/10	0/10
FM-Force	7/10	0/10	0/10	0/10	7/10	0/10
Trajectory Transfer	10/10	9/10	7/10	0/10	8/10	0/10
Ours	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**

Table 2. Success rates of different approaches on the puzzle insertion tasks. By extracting parameterized EC primitives from a single augmented demonstration and executing compliant motions within the contact space, our approach consistently outperforms all baseline methods and demonstrates efficient generalization to unseen puzzle geometries.

identify the EC types and contact-changing events for in-contact phases, and builds a policy composed of three EC primitives. The demonstration and augmentation process are illustrated in Extension 4.

We evaluate our policy on Lock 1 under two object poses with 10 trials each. A trial is defined as a success if the door lock is toggled. As seen in Table 3 and Extension 5, our approach succeeded in all 20 trials. Notably, the robot was unable to insert the key by simply replaying the free-space and plane trajectories. Instead, the in-built exploration actions of the plane primitive, e.g., visual servoing and spiral search, are necessary for the successful insertion across all trials.

These results clearly demonstrate that tracking Cartesian positional trajectories is prone to failure for fine-grained contact-rich tasks. By contrast, our approach utilizes EC primitives to exploit contact feedback, yielding robust execution.

We then evaluate this policy on Lock 2. As shown in Table 3, direct application of the policy fails to insert the key. This failure is primarily caused by two factors. First, differences in object appearance lead to errors in the visual alignment. Second, the increased height of Lock 2 causes the robot to make premature contact with the surface, terminating the free-space primitive at an incorrect pose and propagating errors to the subsequent plane primitive. These instance-specific differences are not captured by our EC primitives. As a result, the policy fails to open Lock 2.

We address these failures using targeted human corrections (Extension 6). After executing the plane primitive, the robot fails to observe the expected breaking-contact event that indicates successful insertion. This discrepancy is correctly recognized as an execution failure, triggering a request for human correction. We then provide a correction that guides the end-effector from the failed pose toward the keyhole. This correction is instantiated as an additional plane primitive with an updated demonstration trajectory and a new reference image for visual servoing. Evaluating the refined policy on Lock 2 yields improved success rates of 7/10 and 8/10 for Pose 1 and Pose 2, respectively.

The remaining failures arise from collisions between the key and the hole edge, which trigger an unexpected making-contact event and is then recognized as an execution failure. We then provide a second correction that introduces a wiggling motion, tilting and rotating the key around the hole, to resolve rotational misalignment. With this second

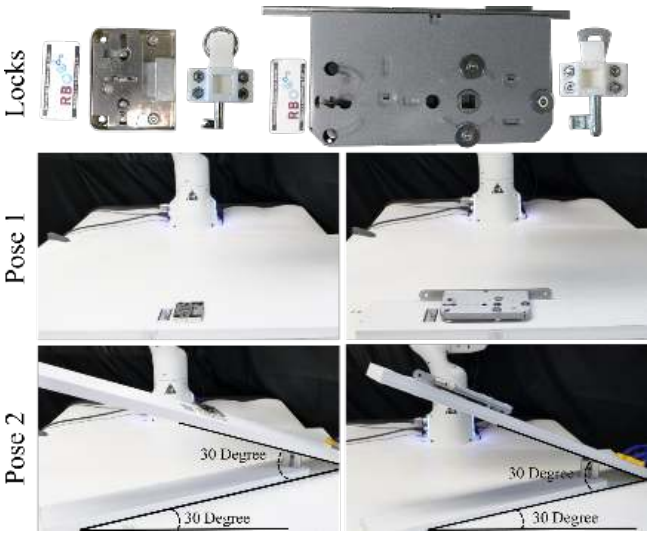


Figure 11. Experimental setup for the door-lock manipulation task. The policy extracted from Lock 1 is extended to Lock 2 via two targeted corrective demonstrations. We intentionally tilt the mounting surface to demonstrate that our approach does not rely on a tabletop assumption.

correction, the policy succeeds in all 20 trials (see Extension 6).

This experiment underscores the critical importance of incorporating human corrections to manage out-of-distribution instance variations that are difficult to anticipate and model in the policy representation. Our approach detects such failures by monitoring contact transitions. Moreover, the modular and interpretable EC-based policy structure supports seamless integration of corrective demonstrations, enabling targeted adaptation to new task configurations without extensive retraining.

Number of corrections	Lock 1		Lock 2	
	Pose 1	Pose 2	Pose 1	Pose 2
No correction	10/10	10/10	0/10	0/10
1st correction	-	-	7/10	8/10
2nd correction	-	-	10/10	10/10

Table 3. Success rates for manipulating Lock 1 and Lock 2 under different numbers of human corrections.

5.5 Coffee Machine

We next evaluate our approach on a coffee machine manipulation task to demonstrate that online adaptation contributes to generalization to imprecise EC models. A policy composed of three EC primitives is constructed from a single augmented demonstration, as shown in Fig. 13. As shown in Extension 7, the policy reliably reproduces the demonstration across 10/10 trials, even under pose variations of the machine.

Next, we introduce modeling errors into the EC model of the rotation primitive, shifting the rotation axis by 20° and 10 cm from the rotation center, as visualized in Fig. 14. Our approach successfully corrects these errors by generating compliant motions and updating its EC model online based on observed movements (see Extension 8). Consequently, our method completes 9 out of 10 trials, whereas a baseline

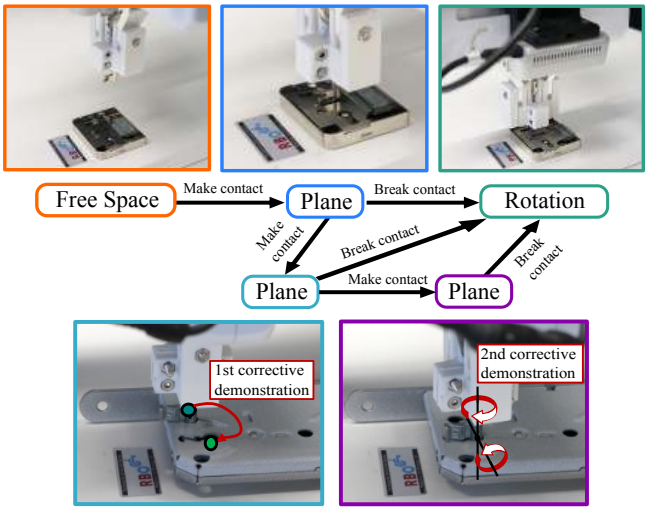


Figure 12. Policy refinement for door lock manipulation. The policy learned from Lock 1 is extended with two corrective demonstrations to handle unseen contact changes in Lock 2. The first correction guides the end-effector from an incorrect pose toward the keyhole, while the second correction introduces wiggling motions to resolve misalignment during insertion. Our EC-based policy representation enables seamless integration of these corrections as additional EC primitives without retraining the full policy.

approach following a precomputed trajectory without online adaptation consistently fails, as the geometric errors rapidly drive the reaction forces beyond safe interaction limits (see Fig. 14).

Overall, our approach reliably completes the task despite pose variations, reduced manipulability near joint limits, and imprecise EC models. It generalizes well because we do not attempt to learn complete EC models from a single demonstration. Instead, our primitives integrate sensory feedback to update models at run time. As we show, this online adaptation is essential for real-world physical interaction; even small modeling errors quickly generate large reaction forces that cause manipulation failures.

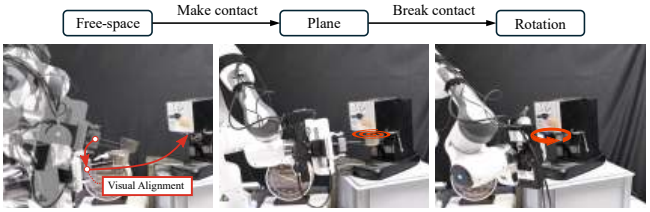


Figure 13. Execution of the coffee machine manipulation policy consisting of three EC primitives. First, the robot performs the visual alignment and executes a transferred trajectory. Upon making contact with the machine, the robot employs a plane primitive to align the portafilter with the insertion slot. Finally, upon detecting a contact-breaking event, it uses a rotation primitive to secure the portafilter.

5.6 Latches

We then test our approach on complex mechanisms involving multiple DOFs. The first such task is latch manipulation, illustrated in Fig. 15. Opening a latch involves a sequence of actions: grasping the knob, sliding it, lifting it until a pin passes a slot, sliding to the end position, and finally lowering it down.

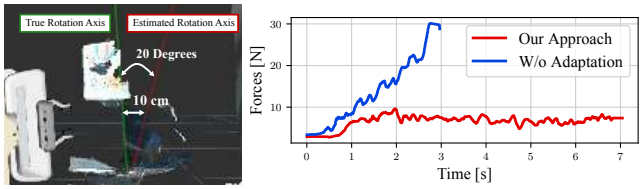


Figure 14. Evaluating policy performance under modeling errors. **Left:** We inject errors into the rotation EC model by applying a translational offset and an angular tilt to the rotation axis. **Right:** Our approach continuously updates the EC parameters based on the observed movement, successfully rotating the portafilter while maintaining low reaction forces. In contrast, executing the task without adaptation quickly exceeds the force limit, resulting in failure.

Method	Latch 1	Latch 2	Latch 3	Latch 4
Our approach	10/10	10/10	10/10	9/10

Table 4. Success rates of grasping and opening latches. The only failure occurred in Latch 4 due to large visual alignment errors that resulted in a failed grasp.

We perform a kinesthetic demonstration on Latch 1 and segment it into five motion phases. After the augmentation, the robot builds a policy that consists of five EC primitives, as shown in Fig. 15. Since all latches have distinct appearances, rather than relying on sophisticated vision algorithms for pose estimation, our approach efficiently extends the existing policy with one grasp demonstration as a correction per latch.

We run 10 trials per latch. For Latch 1–3, success is defined as the pin passing through the slot; for Latch 4, success is defined as the hole in the bar aligning with the protrusions on the plate.

The results are summarized in Table 4 and can be seen in Extension 9. Although each latch requires its own grasping demonstration, the policy extracted from Latch 1 generalizes effectively, achieving 39 successful executions out of 40 trials across all four latches. The single failure was on Latch 4, where its reflective surface disrupted depth measurements, causing a grasp failure. This highlights the need for more robust keypoint extraction and matching algorithms, a limitation we discuss in further detail in Sec. 6.

Our method generalizes to new latches with different geometries by using ECs as the representation and generating motions in the contact space. Crucially, sliding and rotation motions terminate when contact-change events are detected, not when a specific distance or rotation angle is reached. Moreover, when lifting the knob, the translation primitive dynamically adjusts motion direction based on observed movement, allowing the robot to adaptively follow the structure of the constraints. This online adaptation enables generalization to unseen geometric structures and hard-to-model slippages at run time.

5.7 Drawers with Handles

Similar to the latch experiment, the drawer experiment again shows that ECs enable generalization across geometric variations. In this task, the robot must open two drawers with different handle rotation angles (see Fig. 16).

Our approach builds a policy from a kinesthetic demonstration on the drawer with the silver handle. The demonstration is segmented into three motion phases:

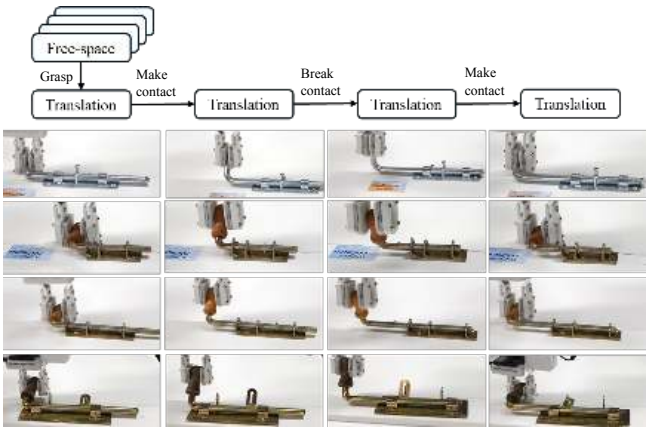


Figure 15. EC-based policy execution across different latches. The top row illustrates the policy structure composed of five EC primitives. The image sequences show successful execution on four different latches using a policy extracted from Latch 1 and adapted only with additional grasp demonstrations. By generating motions in the contact space, the policy generalizes to all latches varied in geometry.

grasping the handle, pushing it downward, and pulling the drawer. After augmentation, we obtain a policy consisting of one free-space primitive for grasping and two translation primitives connected by a breaking-contact event. The handle-rotation behavior is recognized as a translation primitive due to the small observed rotation motion. Nonetheless, the robot can still rotate the handle with a translation primitive by compliantly following the admissible motion direction estimated at run time.

We run 10 trials per drawer and consider a trial successful if the drawer is opened. Each drawer is affixed with a sticker to provide consistent SIFT features for the visual alignment. As shown in Extension 10, our approach succeeds in all 20/20 trials across both drawers, despite differences in required rotation angles and variations in drawer poses.

These results further support our central claim: ECs separate generalizable contact information that captures the task structure while abstracting away task-irrelevant details, such as exact rotation angles. This separation simplifies the learning process and enables sample-efficient generalization to novel instances that vary in task-irrelevant properties.

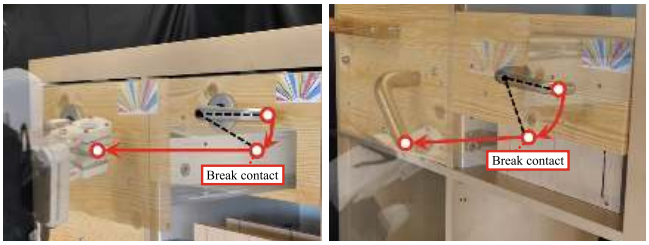


Figure 16. Execution of the policy for opening two drawers. By generating motions that follow the kinematic structure of the handle and drawer, a single policy reliably succeeds despite unseen variations in object pose and geometry.

5.8 Functional Manipulation Benchmark

Finally, we apply our approach to two long-horizon assembly tasks (Luo et al. 2025; Heo et al. 2025) which involve multiple free-space and in-contact motions. As shown below, the complex behaviors required for these tasks can be effectively reproduced using a sequence of EC primitives, parameterized from a single augmented demonstration.

We start with the single-object insertion task from the Functional Manipulation Benchmark (Luo et al. 2025). The robot needs to grasp the object, reposition it on a fixture for a regrasp, and finally insert it into the board. We provide a demonstration using the square-circle object. The demonstration is segmented into four free-space motion phases and two in-contact motion phases. After augmentation, we obtain a policy composed of six EC primitives, as shown in Fig. 19a.

We test the policy in 10 trials using the same square-circle object, with varied poses and tilts of the plate. A trial is successful if the object is fully inserted into the board. The policy succeeds in 9 out of 10 trials. The one failure is due to misalignment, causing the object to collide with the board and trigger a making-contact event. The robot detects this by monitoring contact-change discrepancies. We then provide a corrective demonstration involving a wiggling motion to resolve the misalignment, similar to the one used in the door lock task (Sec. 5.4). With this correction, the policy succeeds in all 10 trials, as shown in Table 5 and Extension 11.

The policy with the square-circle object can be transferred to the oval object with two additional human corrections (see Fig. 17). When applying the policy to the oval object, the robot fails to detect a breaking-contact event after executing the plane primitive, because the oval object does not fit the square hole. A second corrective demonstration is then provided to guide the oval object toward the correct hole. With this correction, the policy succeeds in 3/10 trials.

The remaining failures are due to lost grasps during execution. The grasp pose from the square-circle object does not transfer well to the oval object due to differences in their shape and thickness, leading to failures of a lost grasp. These failures are detected by monitoring the gripper state during the execution. We introduce a third correction that grasps the oval object. With both the second and third corrections applied, the policy succeeds in 9 out of 10 trials, as can be seen in Extension 12. The one failure is caused by a large in-hand pose error after grasping, which is shown in Fig. 18 and discussed in the limitation Section.

5.9 One-Leg Assembly Task

Our method is also applicable to the FurnitureBench benchmark (Heo et al. 2025). We evaluate it on the one-leg assembly task, where the robot grasps a leg, aligns it with a hole in the board, and screws it in.

Our approach segments a demonstration into five motion phases, followed by an augmentation process to recognize the underlying ECs and detect contact-changing events, and enhance visual features for visual alignment. The resulting policy consists of five EC primitives, as shown in Fig. 19b.

The demonstration, augmentation, and evaluation are shown in Extension 13. Despite substantial variations in object pose, the robot successfully completes 9 out of 10 trials. The single failure is due to a collision between the robot and the insertion board, resulting in an incorrect grasp, as shown in Fig. 18.

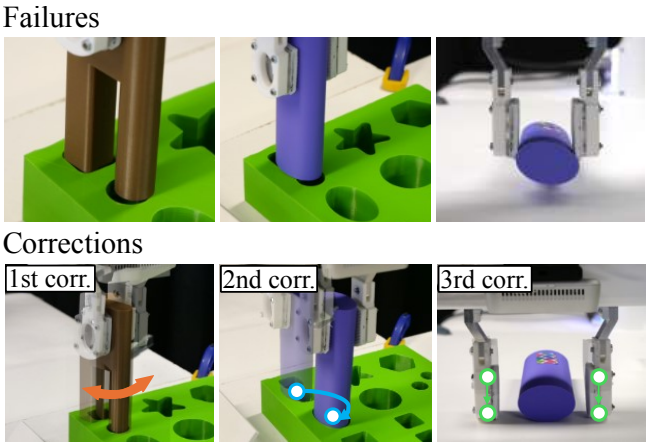


Figure 17. Failures and corrections in the FMB assembly task. The top row shows three failure cases: object-board collision, wrong hole insertion, and lost grasp. The bottom row illustrates the corresponding human corrections: local wiggling to resolve misalignment, retargeting the correct insertion hole, and adjusting the grasp pose.

Objects	No corr.	1st corr.	2nd corr.	3rd corr.
Square-circle	9/10	10/10	-	-
Oval	0/10	-	3/10	9/10

Table 5. Success rates for inserting square-circle and oval objects into the board under different numbers of human corrections.

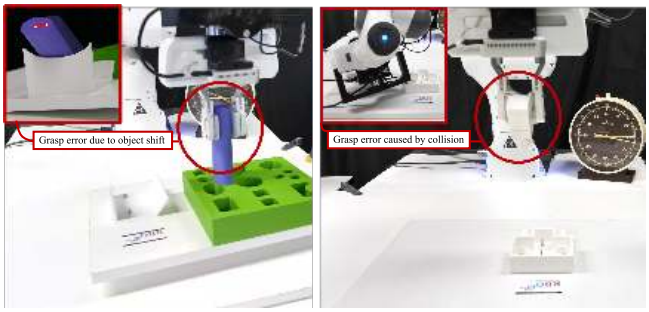


Figure 18. Two failure cases observed in the FMB and FB tasks due to incorrect grasp poses. **Left:** Tilting the plate causes the oval object to shift, leading to an inaccurate grasp. **Right:** During the execution of a free-space primitive, the robot collides with the insertion board, also resulting in errors in the grasp pose. These failures underscore the importance of integrating sensors for in-hand object pose estimation for precise assembly tasks.

5.10 Lessons Learned From Real-World Contact-Rich Manipulation

Based on our extensive real-world experiments, we share several hands-on lessons that reveal the fundamental challenges in contact-rich manipulation. First, exact task trajectories, like the angle needed to turn a handle or open a latch, are nearly impossible to infer from visual observations alone. Second, when manipulating tightly constrained mechanisms, tiny perception or control errors cause massive reaction forces and jamming. These failure states are absent from demonstration data because skilled humans naturally avoid them. Third, even if the robot knows the correct motion and force profile, complex physical interactions, like continuous slippage between the gripper and the object, can still cause trajectory tracking to fail. Finally, in long-horizon tasks like the Functional Manipulation Benchmark and FurnitureBench, these errors compound, and a single failure would already fail the entire sequence. This explains why policies trained end-to-end on hundreds of demonstrations still struggle on these benchmarks (Luo et al. 2025; Heo et al. 2025). Together, these practical realities make it hard to learn a policy that reliably operates on the original demonstration object, let alone generalizes to new task settings.

In contrast, our results show that these complex behaviors are fundamentally *simple*. Following our four-step pipeline, we first leverage a single demonstration and subsequent self-guided augmentation to extract a generalized policy structure based on ECs. We then rely on online adaptation via compliant control to recover instance-specific details and human corrections to handle out-of-distribution failures. Taken together, the aforementioned challenges are effectively resolved by building an EC-based policy through this pipeline. Without ECs, learning these tasks is unnecessarily difficult.

These findings naturally raise a central question: What is the more promising path to generalization, scaling up data or identifying the right inductive biases? We discuss this question in greater depth in Sec. 7.

6 Limitations and Future Work

Although the previous experiments demonstrate that our LfD approach generalizes effectively from a single demonstration per task, several key limitations remain for future work to address.

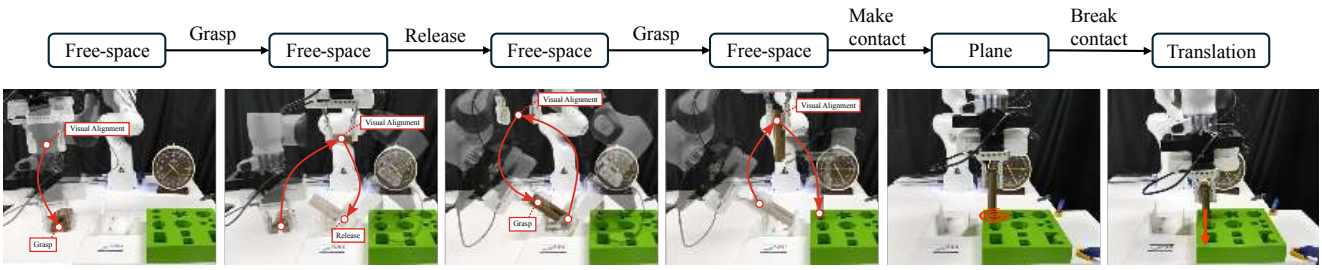
6.1 Application Domains

The strong generalization of our approach is mainly attributed to the EC-based representation. While we have successfully applied this representation to a range of contact-rich manipulation tasks, it remains domain-specific.

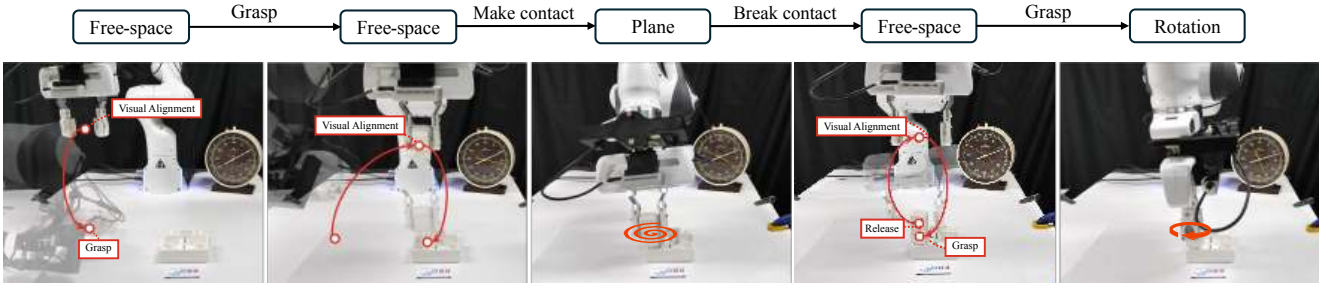
Two main reasons restrict our application domains. First, our primitives focus on geometric constraints rather than exact Cartesian trajectories. This works for many rigid-object manipulation tasks but fails when the specific shape of the motion is the goal, such as in writing or drawing. Second, our system requires discrete changes in the contact state (making- and breaking-contact events) as primitive transitions. While this makes the robot less dependent on exact positions, it prevents the system from handling tasks with continuous contact, for example tasks that require specific wrench values such as peeling, grinding, or wiping a table (Lew et al. 2023; Chen et al. 2024a; Ke et al. 2025), or tasks involving deformable materials such as cloth folding. To broaden applicability, we need to extend our approach with more types of primitives.

Below we discuss which extensions are directly compatible with our architecture and which require more substantial modification.

Our current system implements EC primitives for three major geometric constraints: plane, prismatic, and revolute kinematic constraints. However, it is not limited to these constraint types. The central idea is to instantiate a coarse EC model from a demonstration and then rely on compliant execution to refine the model parameters online through direct interaction with the constraint. In this sense, the



(a) Execution of a single-object insertion policy consists of 6 EC primitives. ① The robot begins with a free-space primitive to grasp the object. ② It then executes a second free-space primitive: it replays part of the demonstration, performs visual realignment to the fixture, and follows the transferred trajectory to place the object on the fixture. ③–④ Two additional free-space primitives are executed to regrasp the object and approach the insertion board. ⑤ Upon establishing contact with the board, the robot performs a plane primitive to align the object with the hole until a breaking-contact event is detected. ⑥ Finally, it transitions to a translation primitive that compliantly inserts the object into the hole.



(b) Execution of a one-leg assembly policy consists of 5 EC primitives. ① The robot starts with visually aligning to the leg and then executes an adapted free-space trajectory to grasp the leg. ② After grasping, it replays part of the demonstration trajectory, performs another visual alignment, and approaches the board to establish contact. ③ The robot executes the plane primitive that aligns the leg with the hole on the board. ④ Upon detecting a breaking-contact event, it transitions to a subsequent free-space primitive to regrasp the leg in a new pose. ⑤ Finally, the robot completes the assembly by executing a rotation primitive to screw the leg into the board.

Figure 19. Complex multi-stage contact-rich manipulation tasks can be decomposed into sequential motions. Our EC primitives are able to reproduce these EC-based motions from only one demonstration.

same pipeline can be extended to other geometric constraint types, such as spherical, cylindrical, or screw constraints, as long as suitable constraint models are available and their parameters can be fit from observed motion and contact forces (Subramani et al. 2018; Overbeek et al. 2025a; Calinon 2026). Importantly, our four-stage pipeline already exposes the main integration challenges and provides corresponding mechanisms to address them: augmentation is used to disambiguate the constraint type and expose the relevant contact transitions, while corrective demonstrations can add missing object-specific details when the original coarse model is insufficient.

Moreover, we should incorporate trajectory-centric primitives, which extract task-relevant motion and force characteristics from demonstrations (Saveriano et al. 2023; Kober et al. 2015; Ugur and Girgin 2020; Wang et al. 2025b; Calinon 2016; Zhou et al. 2016; Li et al. 2023a; Zhou et al. 2019). This allows our system to be extended to non-geometric constraint tasks such as wiping a table. In addition, it is desired to include primitives based on end-to-end methods as fallback mechanisms when the predefined primitives are not suited for a given task (Luo et al. 2024; Xue et al. 2025).

Importantly, incorporating non-geometric primitives would require further modifications to our system. Specifically, our contact-based augmentation strategy and funneling strategies are tailored to geometric-constraint primitives: they deliberately apply forces to establish or maintain contact, allowing the system to disambiguate the EC primitive corresponding to each motion phase in the demonstration and to expose the desired contact-changing events that trigger primitive transitions. These strategies do not apply to more complex contact transitions or to contact-free primitives. A promising direction is to use vision foundation models that support primitive recognition and selection (Chen et al. 2024b; Tang et al. 2025b; Zhu et al. 2026). Recent advances in code-as-policy systems complement this direction: our existing EC primitives could serve as callable modules that enable agent-generated policies to perform contact-rich interactions with diverse rigid objects. Note that our EC primitives can operate from coarse parameter initializations and refine their parameters online through compliant interaction, thereby providing compact interfaces for high-level policy composition (Liang et al. 2023; Fu et al. 2026; Xiao et al. 2026). Another direction is to use reinforcement learning (RL), both to disambiguate the transitions between primitives (Sharma and Kroemer 2022) and to identify the actions that best expose the desired transition events (Yin et al. 2026).

6.2 Technical Limitations

In addition to application domains, our approach has several limitations in the technical implementation.

First, we assume the demonstration can be segmented into a sequence of ECs. However, our segmentation can fail if the demonstration contains exploratory motions that do not follow any particular EC. We need an interactive segmentation algorithm in which the robot refines segment boundaries using the concept of augmentation and human

corrections. This will improve generalization across different demonstrators.

Second, our free-space primitive is built on a visual alignment process that relies on SIFT-based feature extraction and matching. This primitive fails to extract reliable features on textureless objects and does not generalize across objects with different appearances. We plan to extend the free-space primitive by leveraging alternative keypoint-extraction methods. For example, shape-based keypoints could improve generalization across objects with similar geometries (Zhang and Boularias 2024; Huang et al. 2025; Wang et al. 2025a; Dreczkowski et al. 2025). Importantly, recent visual foundation models have shown promising results in providing dense, geometrically consistent semantic keypoints (Radford et al. 2021; Oquab et al. 2024; Siméoni et al. 2025). Using such foundation models as tools for feature extraction and matching would enable visual alignment across unseen objects, an approach that has shown promising results in recent works (Wang et al. 2024; Di Palo and Johns 2024; Fang et al. 2025; Zhu et al. 2024; Tang et al. 2025a; Seker et al. 2025). Notably, since these feature-extraction and matching algorithms are learnable, they can be fine-tuned with our vision-based augmentation pipeline.

Third, failure cases observed in both benchmarks indicate that our system is prone to errors in unstructured environments. One important reason is that our free-space primitive simply replays the transferred demonstration trajectory and lacks the ability to avoid collisions, adapt to dynamic objects and disturbances that perturb the grasp pose (see Fig. 18). Future work should integrate real-time collision-avoidance and active pose-tracking modules into the free-space primitive, allowing it to adapt the demonstrated trajectory in response to obstacles (Vosylius and Johns 2023; Osa et al. 2017; Cai et al. 2024a; Shaw et al. 2023) and to changes in the object pose during execution (Wen et al. 2024; Liu et al. 2025), making the system more robust in unstructured environments. Moreover, our system should incorporate in-hand object pose estimation modules that allow it to detect deviations between the demonstrated and observed grasp poses (Bauza et al. 2023; Yang et al. 2024; Royo-Miquel et al. 2023; Luo et al. 2017; Mack et al. 2025). A dedicated primitive, either hand-designed (Chen et al. 2025b) or learned (Pitz et al. 2024; Sieler and Brock 2023), that minimizes the grasp-pose error would then enable the subsequent EC primitives to execute successfully.

Fourth, our system requires parameter tuning for specific objects, as shown in Table 6. For example, we need to increase the wrench limit as latches involve higher friction than other objects. Similarly, drawer handles have internal springs and thus require higher pulling forces. For door locks, the threshold for detecting breaking-contact events must be reduced to accommodate the tighter mechanical tolerances. Appendix B provides a more detailed sensitivity analysis of these parameters and shows how the policy degrades when they are not identified correctly.

While these parameters are manually specified in the current implementation due to the limited scope of this work, they can be recovered from our LfD pipeline. The required contact forces and the thresholds for contact-changing events can be directly estimated from the demonstrated motion and force trajectory. Augmentation can further refine these estimates through additional trial-and-error interactions that expose the contact forces and displacements associated with successful and failed contact transitions (Johannsmeier et al. 2019; Beltran-Hernandez et al. 2020; Vuong et al. 2021). Furthermore, when transferring to a new object, corrective demonstrations can provide additional object-specific information through the force and motion trajectories of targeted corrections on that object (Franzese et al. 2021; Peternel et al. 2018). Formalizing this parameter-identification process within the learning pipeline would reduce manual effort and improve generalization to new objects.

Table 6. Changed parameters for manipulating various objects

Param.	Value			
	Door Locks	Coffee Machine	Latches	Drawers
F_m	[5, 5, 5, 2, 2, 2]	[5, 5, 5, 2, 2, 2]	[10, 10, 10, 2, 2, 2]	[25, 25, 25, 2, 2, 2]
MCE^p	5 N	8 N	5 N	5 N
BCE	0.7 cm	0.8 cm	1.5 cm	2 cm

7 Discussion: Inductive Biases and Data Scaling for Generalization

Our real-world experiments show that an interactive LfD system built from simple components can achieve sample-efficient generalization across multi-stage rigid-object manipulation tasks. The key to achieving this generalization is to identify the shared interaction structure—free-space motions and in-contact motions governed by planar, prismatic, or revolute geometric constraints—and represent each task as a sequence of environmental constraint exploitations. We acknowledge that the use of this task-specific inductive bias limits the application domain of the current system (see Sec. 6). Within this scope, however, our experiments show that a single augmented demonstration, supplemented by a few targeted corrections where needed, is sufficient to generalize across the tested task variations more effectively than policy classes that rely on more sophisticated models and larger demonstration sets (Luo et al. 2025; Heo et al. 2025; Ankile et al. 2025; TRI LBM Team 2025). More broadly, our results demonstrate that leveraging a suitable inductive bias, if one can be identified, can substantially improve generalization and data efficiency.

An alternative strategy is to avoid introducing such task-specific inductive biases with the goal of obtaining policies that apply across many different tasks. This approach has been especially successful in natural language processing (OpenAI 2023; Touvron et al. 2023; Bai et al. 2023) and computer vision (Radford et al. 2021; He et al. 2022; Kirillov et al. 2023; Oquab et al. 2024), and it is frequently argued that data scaling will eventually also solve generalization in robotics (Amato et al. 2025).

In this section, we go beyond these experimental results and share our broader perspective on whether data scaling alone will solve robotics. Specifically, we discuss how the problems encountered in robotics differ from those in natural language processing and computer vision. We therefore have little justification for expecting that the game-changing

results from those domains will transfer directly to robotics. Robotics may in fact represent one of the most challenging application domains for such learning methods.

There are at least two reasons that make learning contact-rich manipulation difficult. First, the function we might attempt to learn is basically chaotic, making it very challenging to learn or to model. When considering contacts, small changes to initial conditions lead to drastically different outcomes, including drastically different sensory traces. When multiple, interacting contact points are involved, things get worse. There is little hope to learn such a function—this function might not actually "exist" in a stable form as minor changes to environmental conditions can lead to drastic changes. And even if we could learn this function, the generalization of this function should be rather limited, again, as a result of small changes having *different* drastic consequences not seen during training. In contrast, by building the exploitation of environmental constraints into our approach, all motions seek a contact regime that is easy to maintain and detect. Environmental constraints as an inductive bias eliminate this problem, leading to better generalization.

The second reason is that meaningful training data, direly required for the data-scaling approach, is challenging to come by. This data is most likely to be generated by humans. But humans have such extensive manipulation experience, they are so skilled at such tasks, that the training data we can obtain by observing them is heavily shifted towards small regions of the action space. No matter how many data points (demonstrations) we obtain, it seems highly unlikely that we can cover in any meaningful way those parts of the state space that a robot might encounter during policy execution under different environmental conditions. Again, environmental constraints as an inductive bias resolve this issue by abstracting away task-irrelevant details (e.g., exact motion and force profiles). With this abstraction, we represent what used to be a large region of the state space as a clearly identifiable lower-dimensional manifold. This is valid because the entire original region represents an equivalence class *with respect to the environmental constraint*. This abstraction makes it easy to gather sufficient training data to cover the task-relevant space (one human demonstration augmented by the robot is sufficient in many cases).

While environmental constraints solve these two challenges effectively, one might still wonder whether such inductive biases could be automatically extracted from data. We consider this unlikely because several recent studies show that successful task completion by end-to-end manipulation policies often reflects interpolation within the training data rather than meaningful generalization (He et al. 2026; Fei et al. 2025). The same is true in natural language processing and computer vision, where effective inductive biases—like transformers and CNNs—did not arise from scaling but were intentionally designed to encode symmetry and invariance (Amato et al. 2025).

Importantly, we do not argue that end-to-end methods cannot solve these tasks. Instead, we highlight the benefit of introducing the right inductive bias to achieve sample-efficient learning and generalization across variations within a task domain. While our experiments support this conclusion only for one class of contact-rich tasks—rigid-object manipulation tasks for which the shared interaction structure can be represented by the four patterns considered in this work—we believe that our observations about the properties that make contact-rich manipulation difficult may also apply beyond the specific tasks studied in this paper. In robotics, we have to deal with the "real world," that is, with environments for which no simple models exist and whose dynamics can be chaotic or near-chaotic, as in complex and uncertain contact situations. We therefore believe that it is imperative in robotics to continue to rely on strong and task-specific inductive biases. Learning on top of such biases continues to be a fruitful and desirable path towards more competent robotic systems.

8 Conclusion

We present an interactive LfD system that achieves sample-efficient generalization, from a single augmented demonstration and a few targeted corrections, across various multi-stage contact-rich manipulation tasks with rigid objects. These tasks share a general structure that can be represented as free-space motions and in-contact motions governed by planar, prismatic, or revolute geometric constraints. We leverage this insight to design a system that extracts the task-general structure from a demonstration, augments this representation through self-guided exploration, refines it via human corrections, and recovers instance-specific details online using sensory feedback. Extensive experiments across seven real-world, contact-rich manipulation tasks demonstrate that policies constructed through this four-step pipeline with simple components are substantially more general than those derived from trajectory-centric or purely data-driven methods. Furthermore, based on extensive hands-on experimentation, we provide key insights into why contact-rich manipulation remains difficult to learn and show that environmental constraints serve as a highly effective inductive bias to overcome these challenges. We believe that insights into achieving generalization can inspire future research on designing learning algorithms that exploit environmental constraints in different ways, enabling more principled and general LfD for contact-rich manipulation.

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Supplemental material

This article is accompanied by 15 multimedia extensions. These videos illustrate the incremental policy construction process, visualize the underlying algorithmic mechanisms during execution, and provide continuous recordings of the experimental evaluations. These materials are available online at: https://www.youtube.com/playlist?list=PLDsDQLkZj4vk.

Declaration of conflicting interests

The authors declare that there is no conflict of interest.

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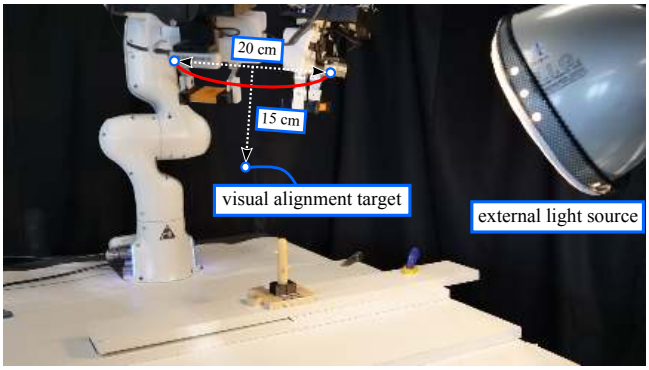


Figure 20. Experimental setup for evaluating the impact of vision-based augmentation under lighting changes. The robot performs visual alignment to return to a target pose, starting from 10 initial poses arranged along a circular trajectory.

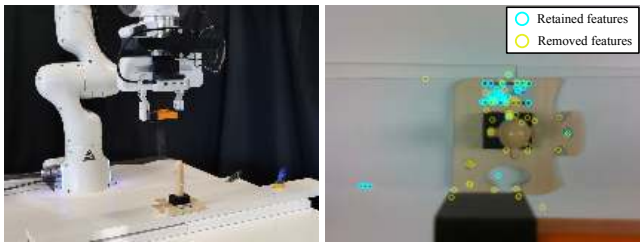


Figure 21. Effect of vision-based augmentation on the insertion puzzle task. **Left:** The robot at the pose where vision-based augmentation is applied. **Right:** Features on textureless regions or object boundaries, which are prone to mismatches or viewpoint-inconsistent, are removed by our vision-based augmentation strategy, leading to a better visual alignment performance.

A Impact of Vision-Based Augmentation

This additional experiment shows how vision-based augmentation and EC exploitation support policy execution under increased visual alignment errors. To do so, we introduce an external light source to change lighting conditions (see Fig. 20).

We begin by moving the robot to a target pose, capturing a reference image, and recording the end-effector pose as ground truth. We then apply vision-based augmentation at this pose to improve image feature quality for the visual alignment (see Fig. 21 and Extension 14). Next, we turn on an external light to change the lighting conditions and test whether the robot can return to the target pose. We define 10 initial poses in a circular path 15 cm above the target and attempt alignment from each. We then compute translational and rotational errors of the end-effector pose relative to the ground truth. As shown in Fig. 22, alignment errors increase significantly when vision-based augmentation is omitted. These results confirm that the vision-based augmentation strategy enhances the visual alignment performance.

We next examine how increased alignment errors affect policy execution. We evaluate two methods that rely on visual alignment: Trajectory Transfer and our approach. Both methods use the same policies from the previous experiment, but are now tested under a modified task setup with changes in object pose and lighting conditions.

As shown in Table 7, Trajectory Transfer succeeds in only 3 of 10 trials without vision-based augmentation due to its sensitivity to visual alignment errors. We observed

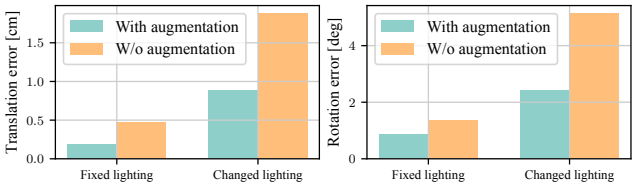


Figure 22. Comparison of average translational (left) and rotational (right) alignment errors in changed lighting conditions. The relatively small increase in error when lighting changes demonstrates the effectiveness of our vision-based augmentation strategy in enhancing the robustness of visual alignment to unseen lighting conditions.

Method	W/o augmentation	With augmentation
Trajectory Transfer	3/10	6/10
Ours	**9/10**	**10/10**

Table 7. Evaluation of Trajectory Transfer and our approach under changed lighting conditions, in which the visual alignment errors increase (see Fig. 22). Our approach benefits from EC primitives that correct visual alignment errors through contact exploitation, achieving higher success rates. It also shows that vision-based augmentation increases the success rates of both approaches by improving visual alignment performance.

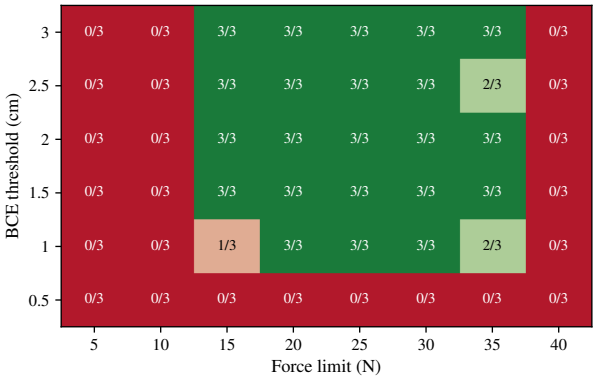


Figure 23. Sensitivity of the drawer policy to object-specific parameters. Each cell reports the number of successful trials out of three for a combination of the force limit (the translational entries of the wrench limit F_m) and the BCE threshold; the cell color interpolates from red $(0/3)$ to green $(3/3)$. The values used in the main experiments (Sec. 5.7) are a force limit of 25 N and a BCE threshold of 2 cm.

that even small orientation errors in the visual alignment can significantly disrupt Trajectory Transfer. With augmentation, success improves to 60% (6/10), consistent with the reduced alignment errors in Fig. 22.

Unlike Trajectory Transfer, our method with vision-based augmentation successfully completes all 10 trials (see Extension 3). This high success rate is attributed to EC exploitations that effectively mitigate visual alignment errors. Specifically, as illustrated in Fig. 6, after executing a free-space primitive, the robot purposefully moves toward the puzzle board to establish contact, thereby reducing positional errors. Then, the plane primitive further compensates for remaining errors by performing visual servoing followed by a spiral search along the surface, ensuring robust execution.

These results further underscore the critical role of incorporating ECs into the learning system. By leveraging ECs, the system can augment demonstrations with additional information and exploit contact feedback to reduce uncertainty, enabling robust execution under varying environmental conditions.

B Sensitivity Analysis of Object-Specific Physical Parameters

As Table 6 shows, our system requires a small number of object-specific physical parameters for certain objects. In this experiment, we analyze how these parameters affect policy performance.

We use the drawer task as the testbed. Its policy depends on two object-specific parameters: the wrench limit F_m and the breaking-contact-event (BCE) threshold. Specifically, the translational entries of F_m determine the maximum force the robot can apply while rotating the handle and pulling the drawer. This force must be large enough to overcome the internal spring and friction of the mechanism, but not so large that the interaction becomes unstable. The BCE threshold determines when the system detects the transition from the EC primitive that rotates the handle to the one that pulls the drawer. It must therefore be large enough to avoid triggering on mechanical play or end-effector slippage, while still being small enough to detect the true breaking-contact event reliably.

We evaluate the same policy used in the drawer experiment in Sec. 5.7. To isolate the in-contact part of the skill, each trial starts from an already grasped pose, so the policy executes only the two EC primitives that rotate the handle and then pull the drawer open. We sweep the force limit over $\{5, 10, 15, 20, 25, 30, 35, 40\}$ N and the BCE threshold over $\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ cm, evaluating all 8×6 combinations. Note that a force limit of 25 N corresponds to translational wrench-limit entries $[25\,\text{N}, 25\,\text{N}, 25\,\text{N}]$ of F_m, while the rotational entries remain fixed at their demonstrated value of 2 N m, as in the other experiments. For each setting, we execute the policy from three initial grasped poses and report the number of successful trials. A trial is considered successful if the drawer is fully opened, consistent with the success criterion in the drawer experiment (see Sec. 5.7).

The results are shown in Fig. 23. We observe that the policy performs reliably for force limits between 15 N and 30 N and BCE thresholds from 1 cm up to at least 3 cm. When the force limit is below 15 N, the robot cannot generate enough force to actuate the mechanism, and the drawer never opens. At a force limit of 35 N, the task remains solvable, but the interaction becomes increasingly aggressive and leads to grasp loss due to slippage. At 40 N, the policy fails because the robot approaches its safety limits. For the BCE threshold, setting it to 0.5 cm causes the breaking-contact event to trigger prematurely due to mechanical play and end-effector slippage, causing the policy to fail consistently.

Overall, this experiment shows that the drawer policy remains reliable over a fairly broad parameter range. This is partly enabled by the compliant controller, which allows us to directly specify meaningful interaction-force limits rather than tuning low-level controller gains, while also adapting the robot motion to the encountered constraints at run time. More importantly, as discussed in Sec. 6, these parameters are physical quantities rather than opaque

tuning knobs. Their initial values can be estimated from the motion and force profiles in the demonstration, while augmentation and corrective demonstrations can provide additional information to refine them. Systematically incorporating this identification process into the pipeline is an important direction for future work.